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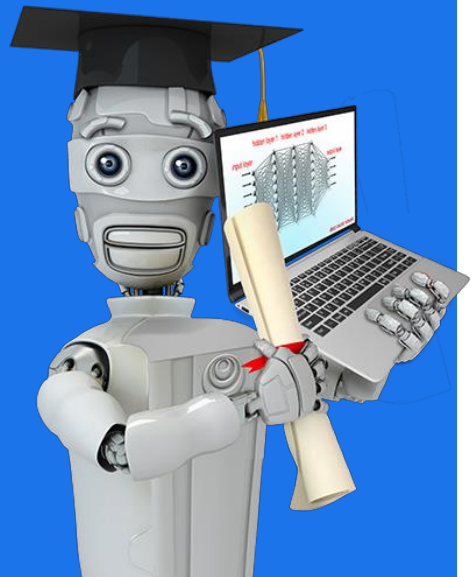
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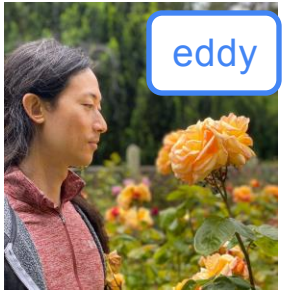
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Machine Learning

Welcome!


eddy


aarti


geoff


Ivy


kin

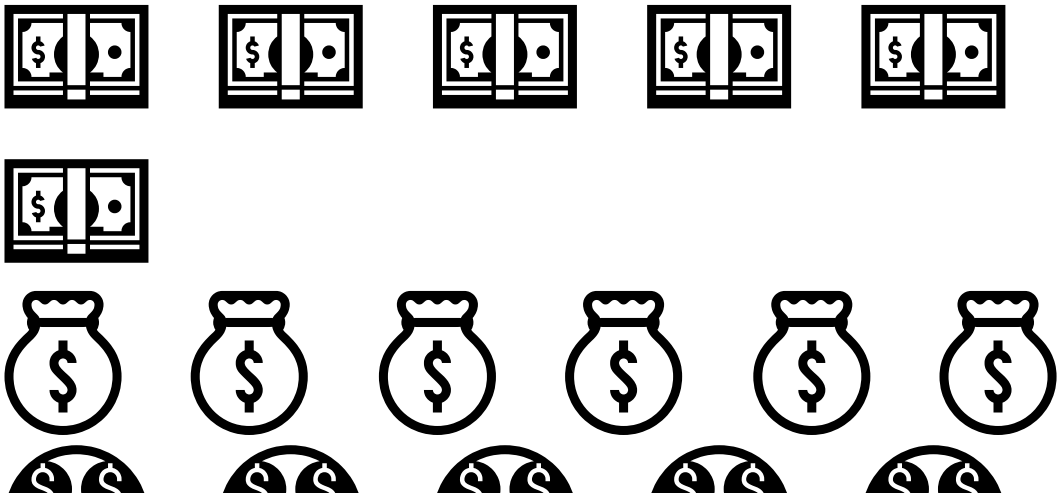

robert


andres


daniel

Re: Urgent Information :) External Spam ×

Congratulations! You've won a million dollars!



Gmail

Compose

Mail

Inbox

Starred

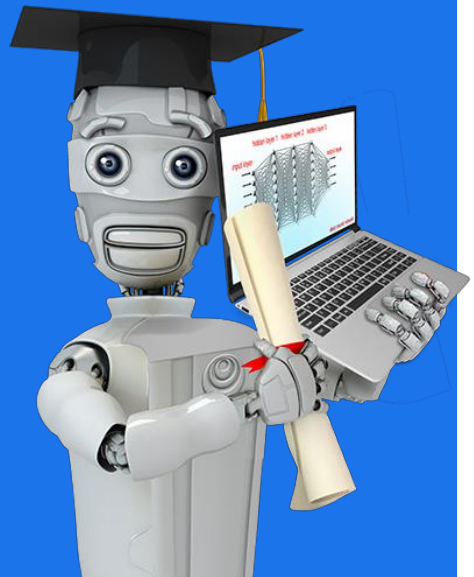
Snoozed

Sent

Drafts 42

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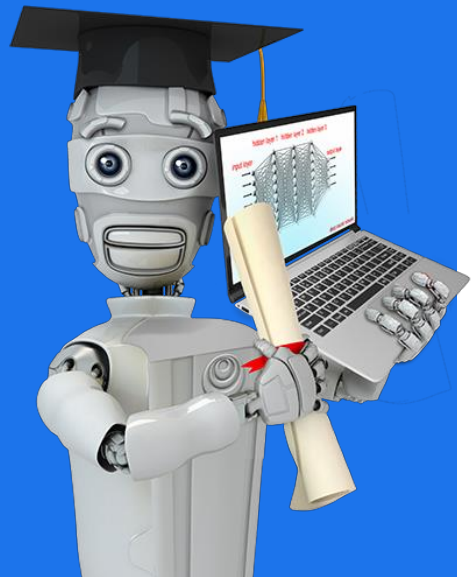


Machine Learning

Applications of Machine Learning

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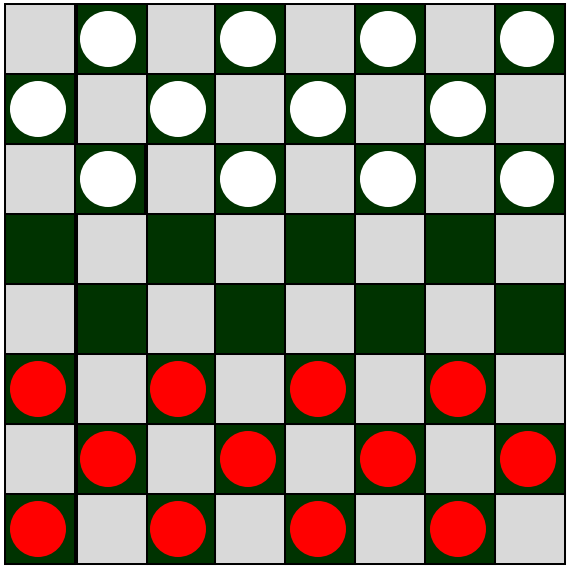
Machine Learning Overview

What is Machine Learning?

Machine learning

"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



Question

If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

- Would have made it better
- Would have made it worse

Machine learning algorithms

rapid advancements

used most in real-world applications

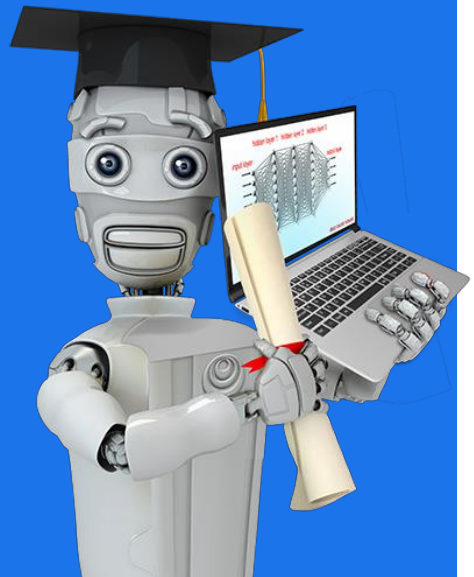
- Supervised learning ← course 1, 2
- Unsupervised learning ← course 3
- Recommender systems
- Reinforcement learning



Practical advice for applying learning algorithms

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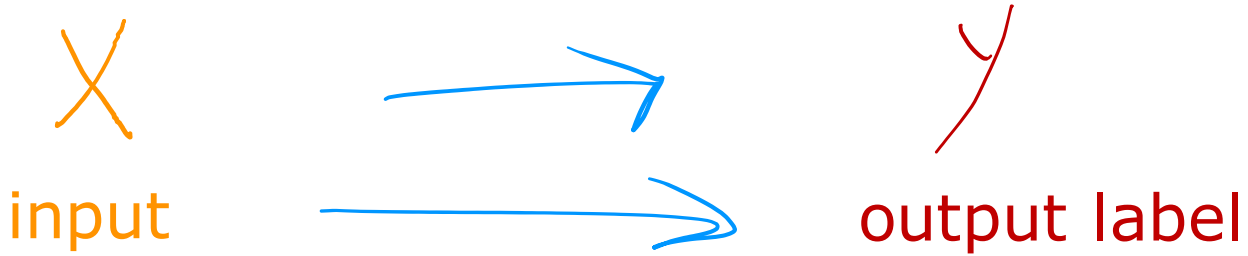
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Machine Learning Overview

Supervised Learning Part 1

Supervised learning

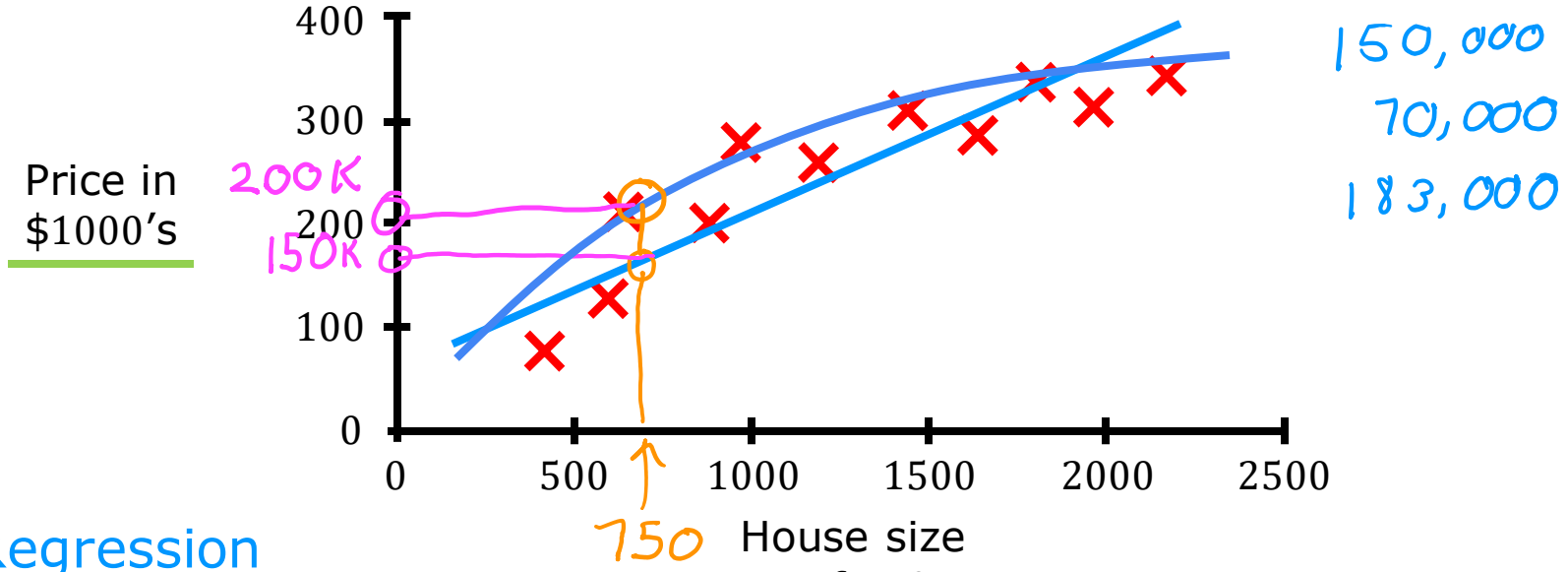


Learns from being given "right answers"

predic

Input (X)	Output (Y)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Spanish	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	self-driving car
image of phone	defect? (0/1)	visual inspection

Regression: Housing price prediction



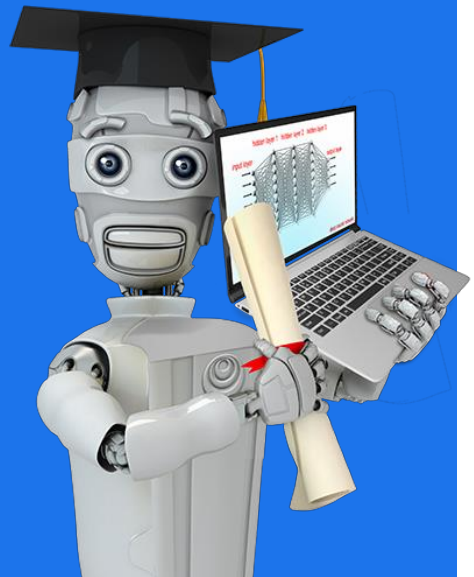
Regression

Predict a number

infinitely many possible outputs

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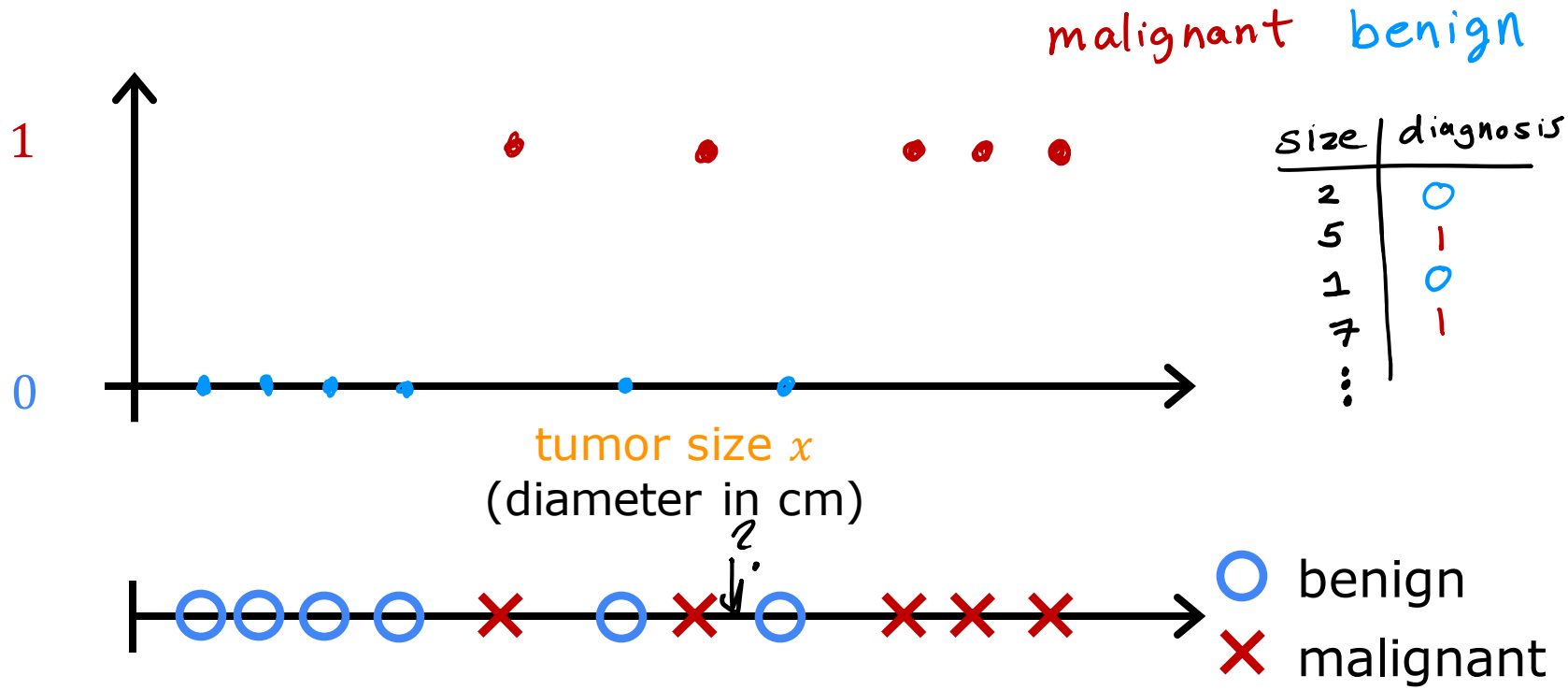
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Machine Learning Overview

Supervised Learning Part 2

Classification: Breast cancer detection

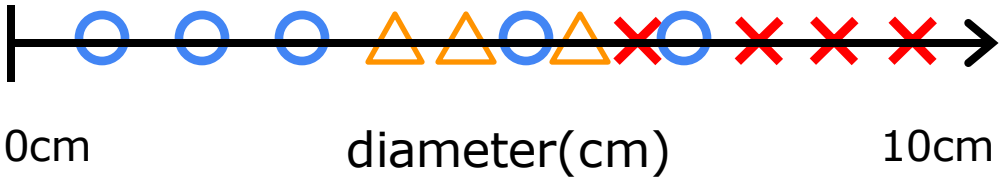


Classification: Breast cancer detection

benign

malignant type1

malignant type 2



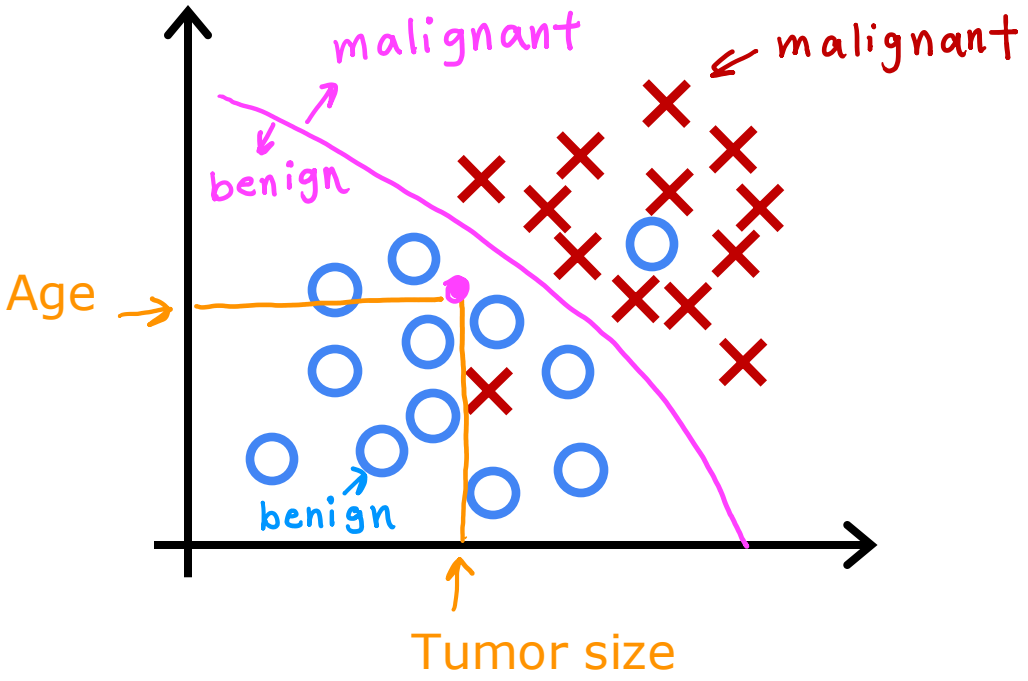
class category

Classification

predict categories cat dog benign malignant 0, 1, 2

small number of possible outputs

Two or more inputs



分割边界

Supervised learning

Learns from being given "right answers"

Regression

Predict a number

infinitely many possible outputs

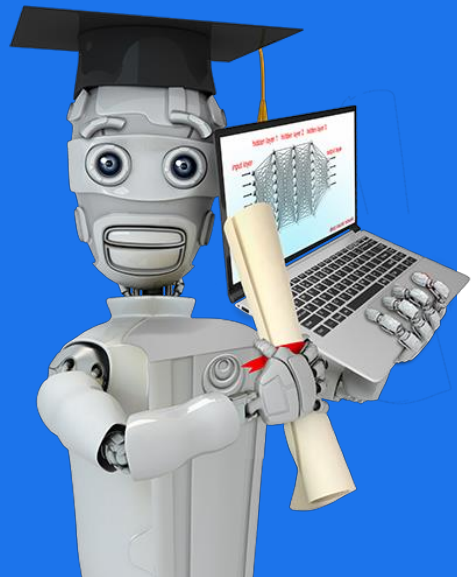
Classification

predict categories

small number of possible outputs

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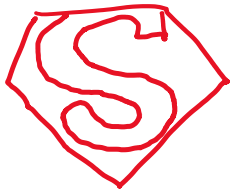


Machine Learning Overview

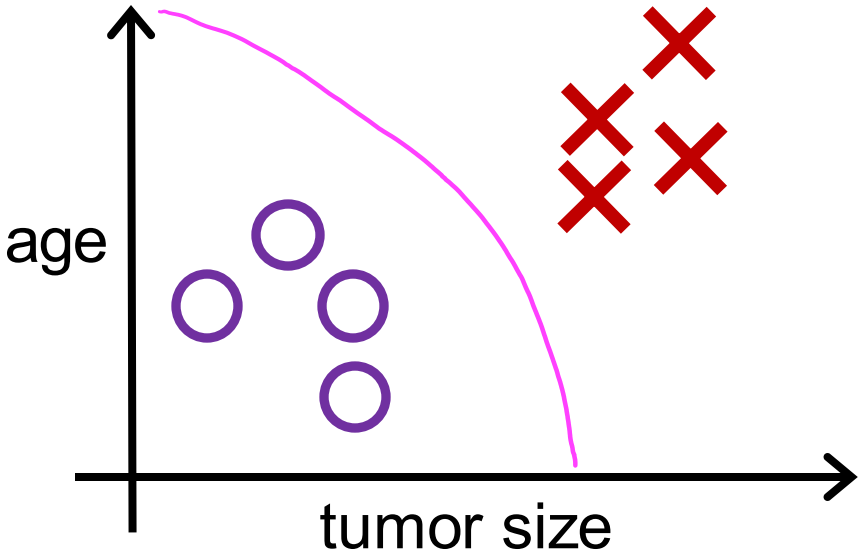
Unsupervised Learning Part 1

Previous: Supervised learning

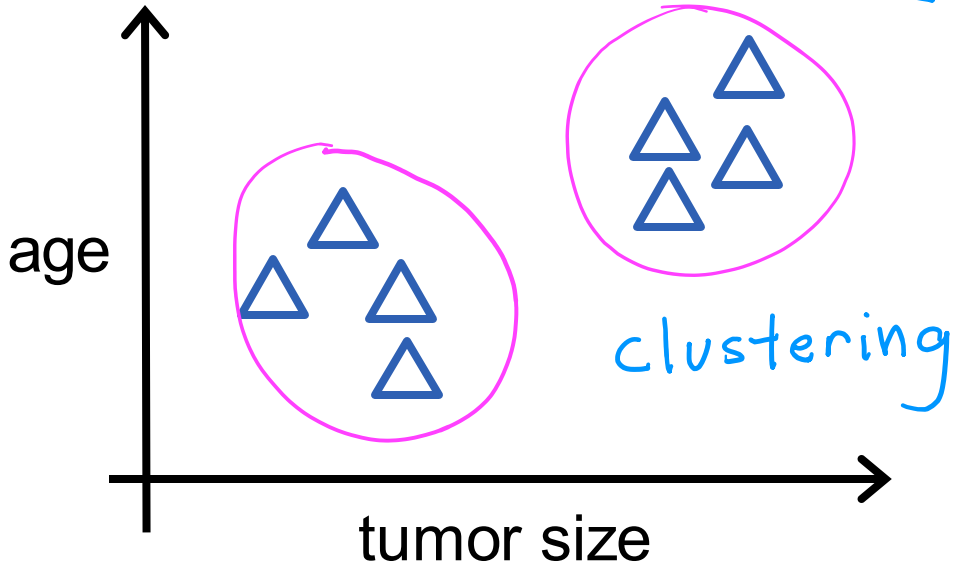
Now: Unsupervised learning



Supervised learning
Learn from data labeled with the "right answers"



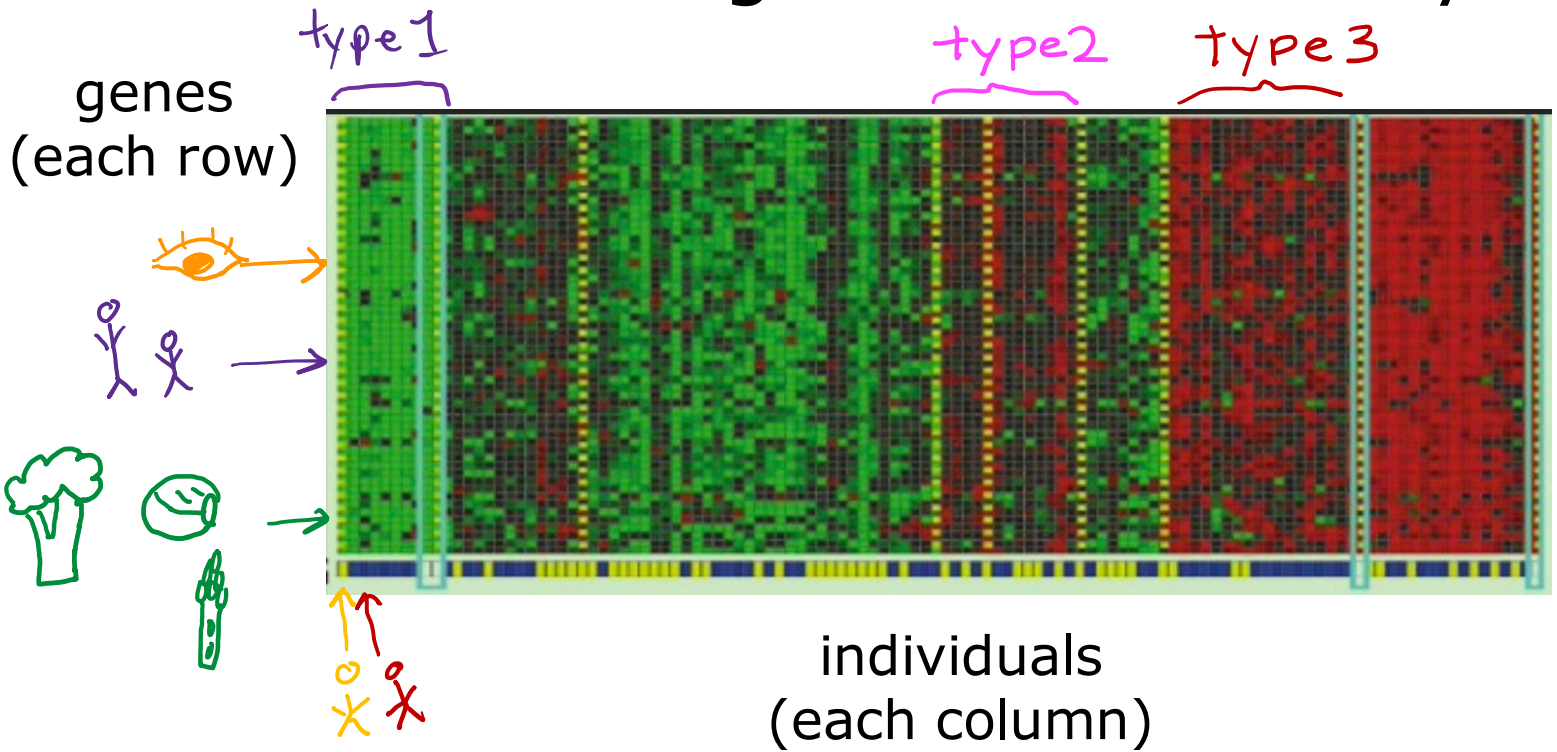
Unsupervised learning
Find something interesting in unlabeled data.



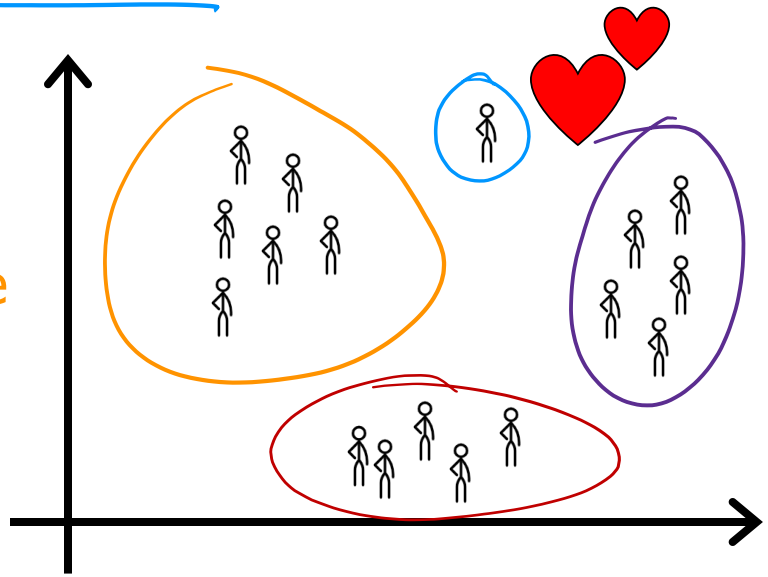
Clustering: Google news



Clustering: DNA microarray



Clustering: Grouping customers



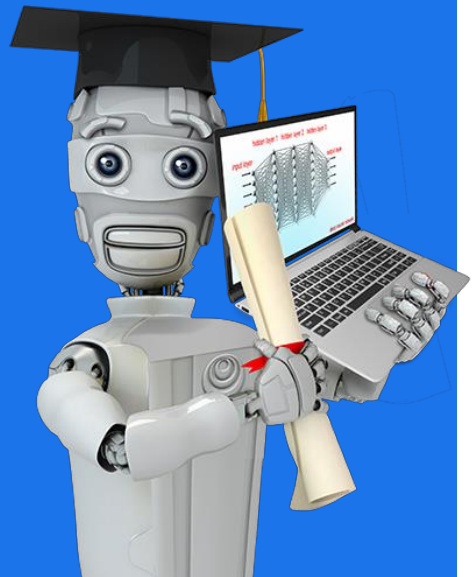
growing skills and knowledge

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Machine Learning Overview

Unsupervised Learning Part 2

Unsupervised learning

Data only comes with inputs x, but not output labels y.
Algorithm has to find structure in the data.

Pattern, feature

降维

<u>Clustering</u>
Group similar data points together.

<u>Dimensionality reduction</u>
Compress data using fewer numbers.

<u>Anomaly detection</u>
Find unusual data points.

OOD

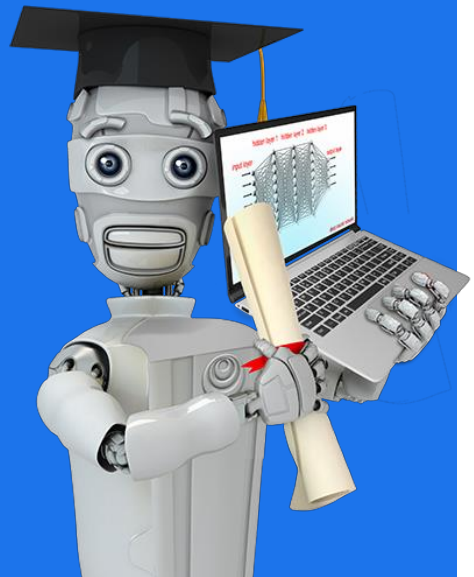
Question

Of the following examples, which would you address using an unsupervised learning algorithm?

- [ ] Given email labeled as spam/not spam, learn a spam filter.
- [x] Given a set of news articles found on the web, group them into sets of articles about the same story.
- [x] Given a database of customer data, automatically discover market segments and group customers into different market segments.
- [ ] Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

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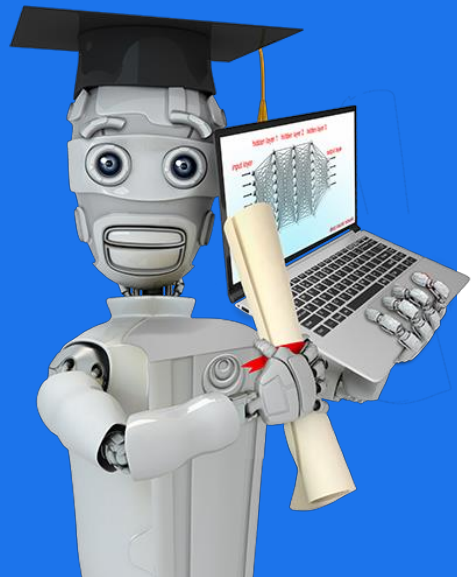


Machine Learning Overview

Jupyter Notebooks

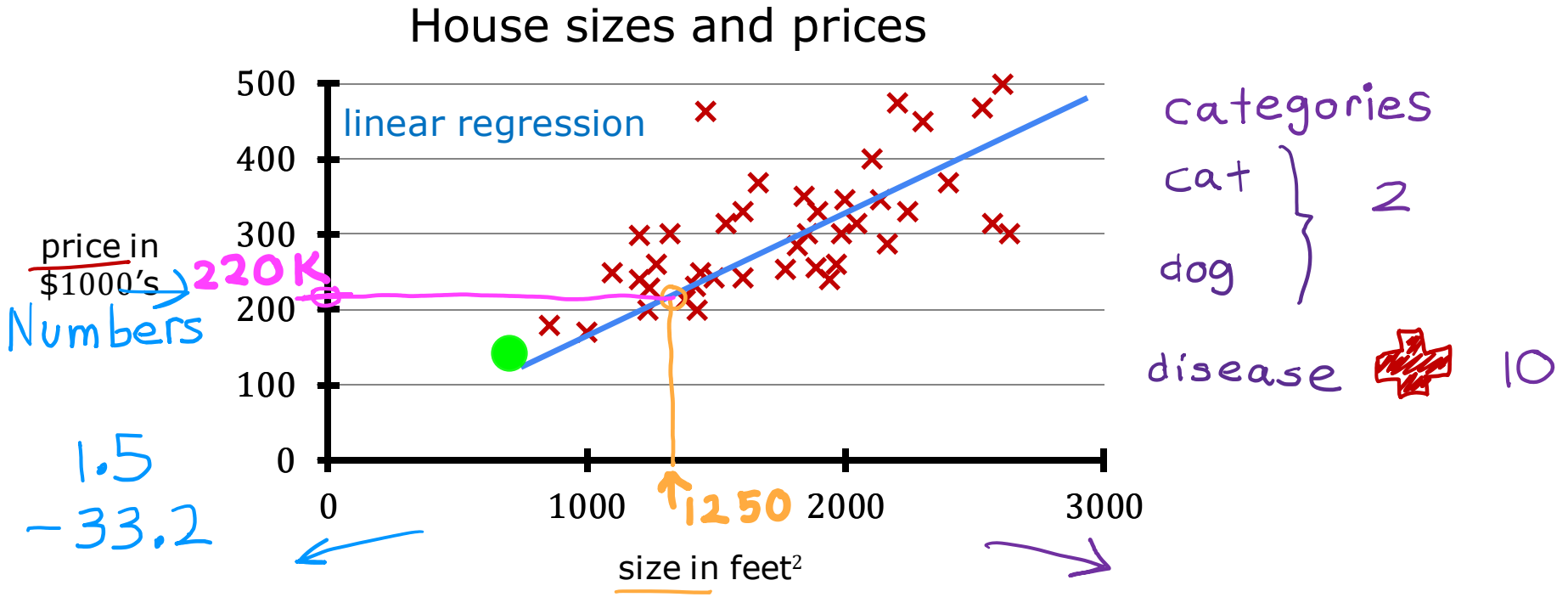
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Linear Regression with One Variable

Linear Regression Model Part 1



Regression model
Predicts numbers
Infinitely many possible outputs

Supervised learning model
Data has "right answers"

Classification model
Predicts categories
Small number of possible outputs

House sizes and prices



Data table

size in feet²	price in $1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Terminology

Training set: Data used to train the model

	x size in feet^2	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
…	…	…
(47)	3210	870

$m = 47$

$x^{(1)} = 2104$ $y^{(1)} = 400$

$(x^{(1)}, y^{(1)}) = (2104, 400)$

$x^{(2)} = 1416$ $y^{(2)} \neq x^2$ not exponent

Notation:

x = "input" variable
feature

y = "output" variable
"target" variable

m = number of training examples

(x, y) = single training example

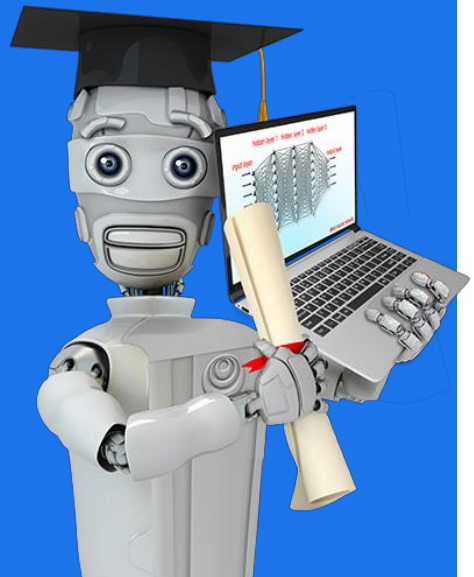
$(x^{(i)}, y^{(i)})$

$(x^{(i)}, y^{(i)})$ = i^{th} training example

index (1st, 2nd, 3rd …)

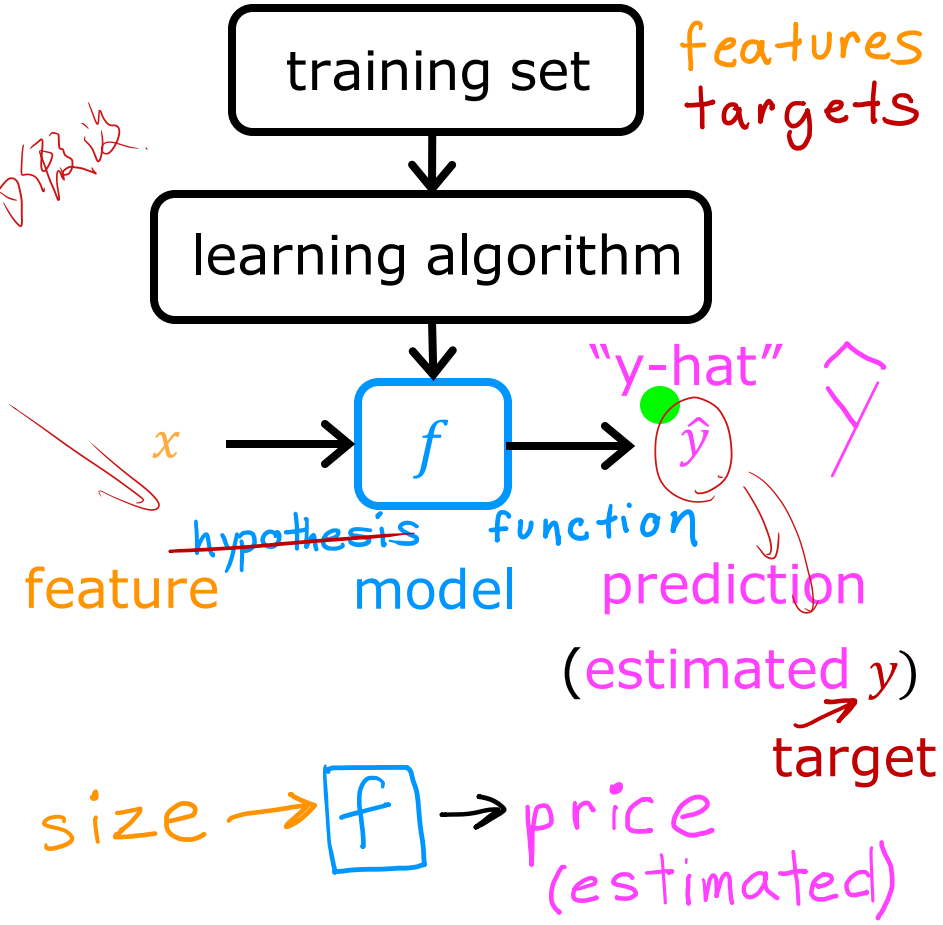
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Linear Regression with One Variable

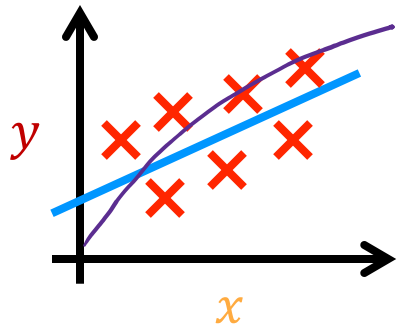
Linear Regression Model Part 2



How to represent f?

$f_{w,b}(x) = wx + b$

$f(x)$



$f_{w,b}(x) = wx + b$

$f(x) = wx + b$

linear

single feature x

Linear regression with one variable.

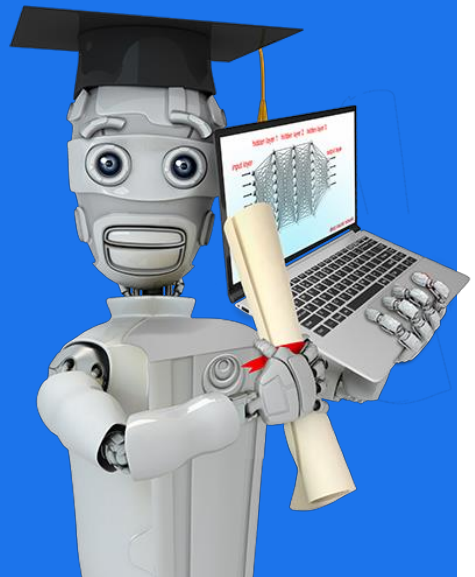
size

Univariate linear regression.

one variable

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Linear Regression with One Variable

Cost Function

Training set

features | targets

size in feet² (x)	price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Model: $f_{w,b}(x) = wx + b$

w, b: parameters

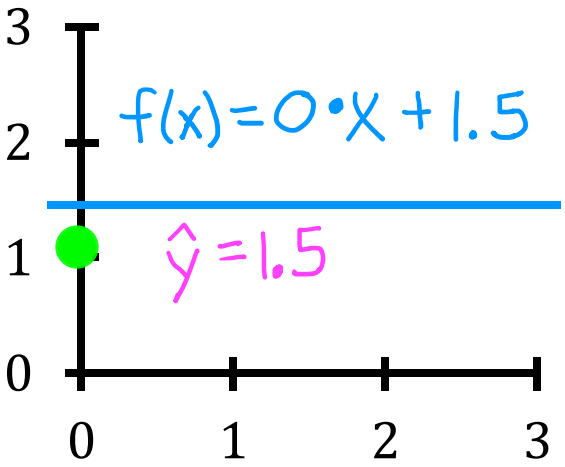
coefficients

weights

What do w, b do?

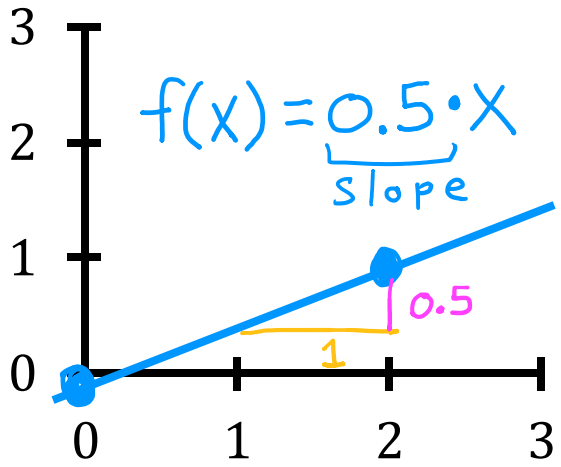
$$f_{w,b}(x) = wx + b$$

$f(x)$



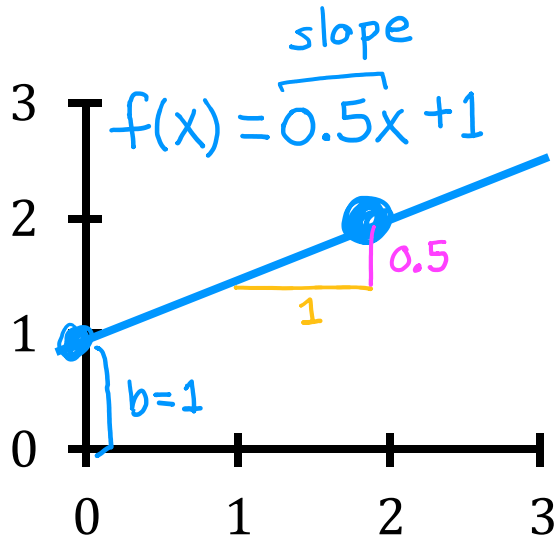
→ $w = 0$

→ $b = 1.5$ (y-intercept)



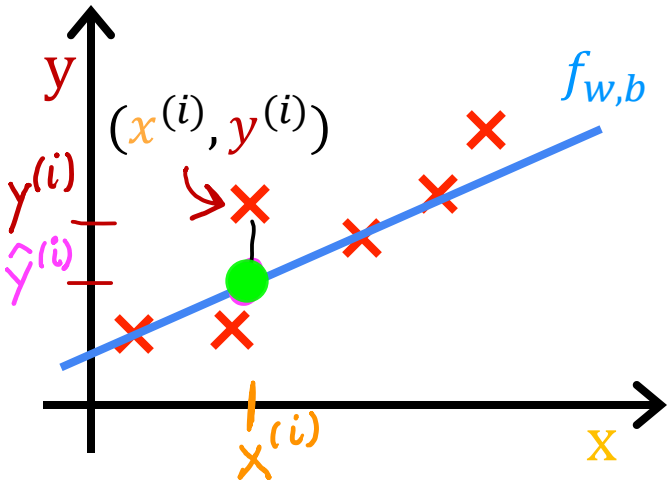
→ $w = 0.5$

→ $b = 0$



→ $w = 0.5$

→ $b = 1$



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)})$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function: Squared error cost function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

$\hat{y}^{(i)}$ − $y^{(i)}$: error

m = number of training examples

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2$$

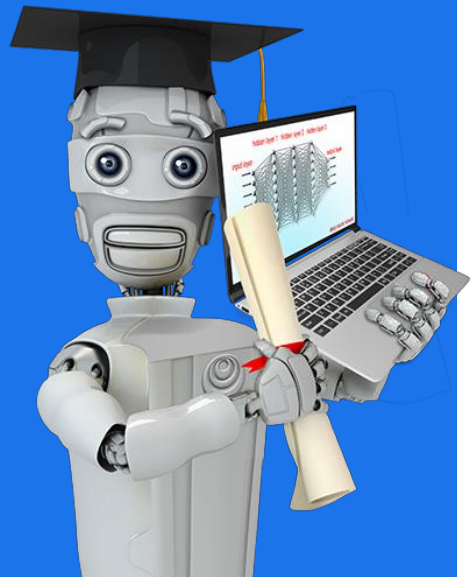
$J(w,b)$: intuition (next!)

Find w, b:

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

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Linear Regression with One Variable

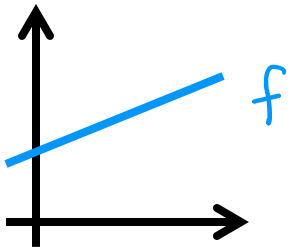
Cost Function Intuition

model:

$$f_{w,b}(x) = wx + b$$

parameters:

w, b



cost function:

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

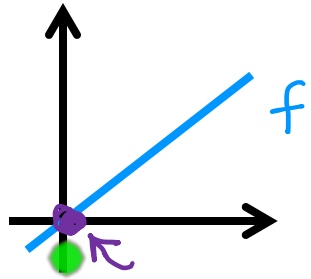
$$\underset{w,b}{\text{minimize}}\, J(w,b)$$

simplified

$$f_w(x) = wx$$

$b = 0$

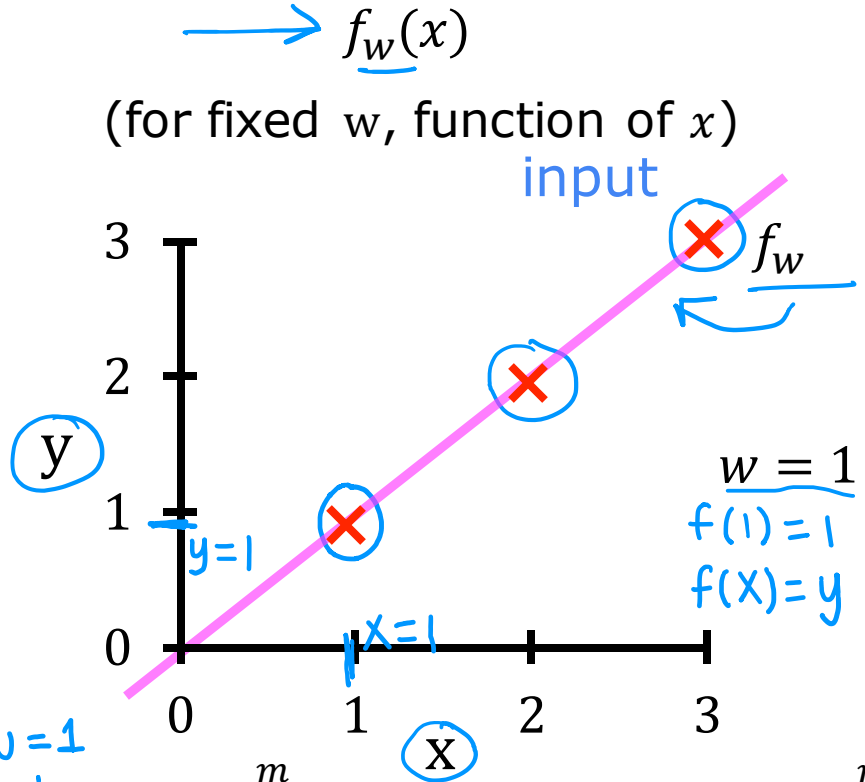
w

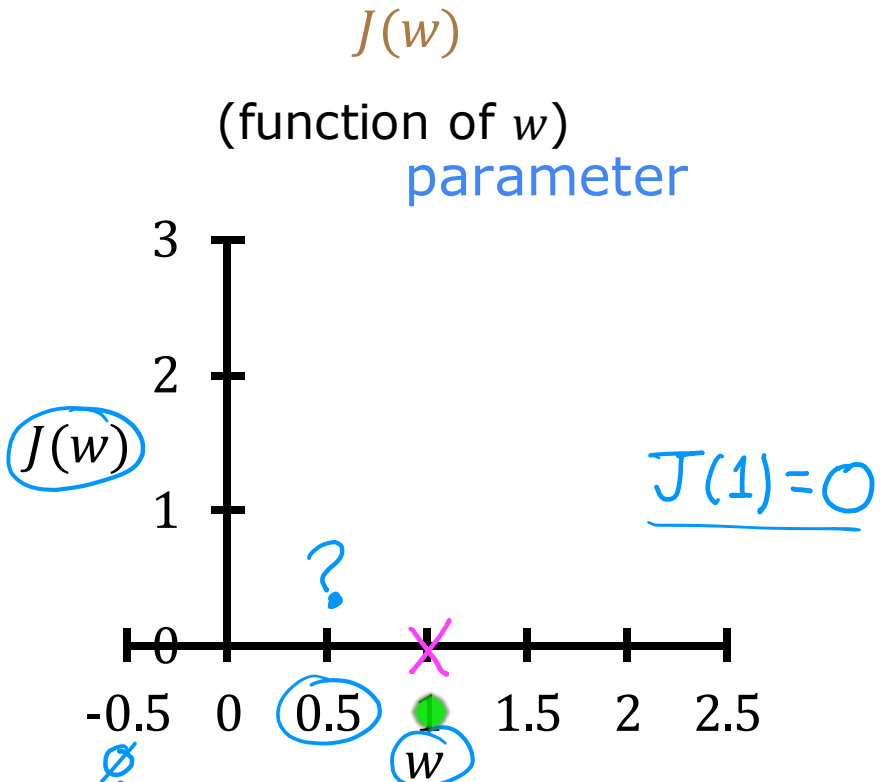


$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}(f_w(x^{(i)}) - y^{(i)})^2$$

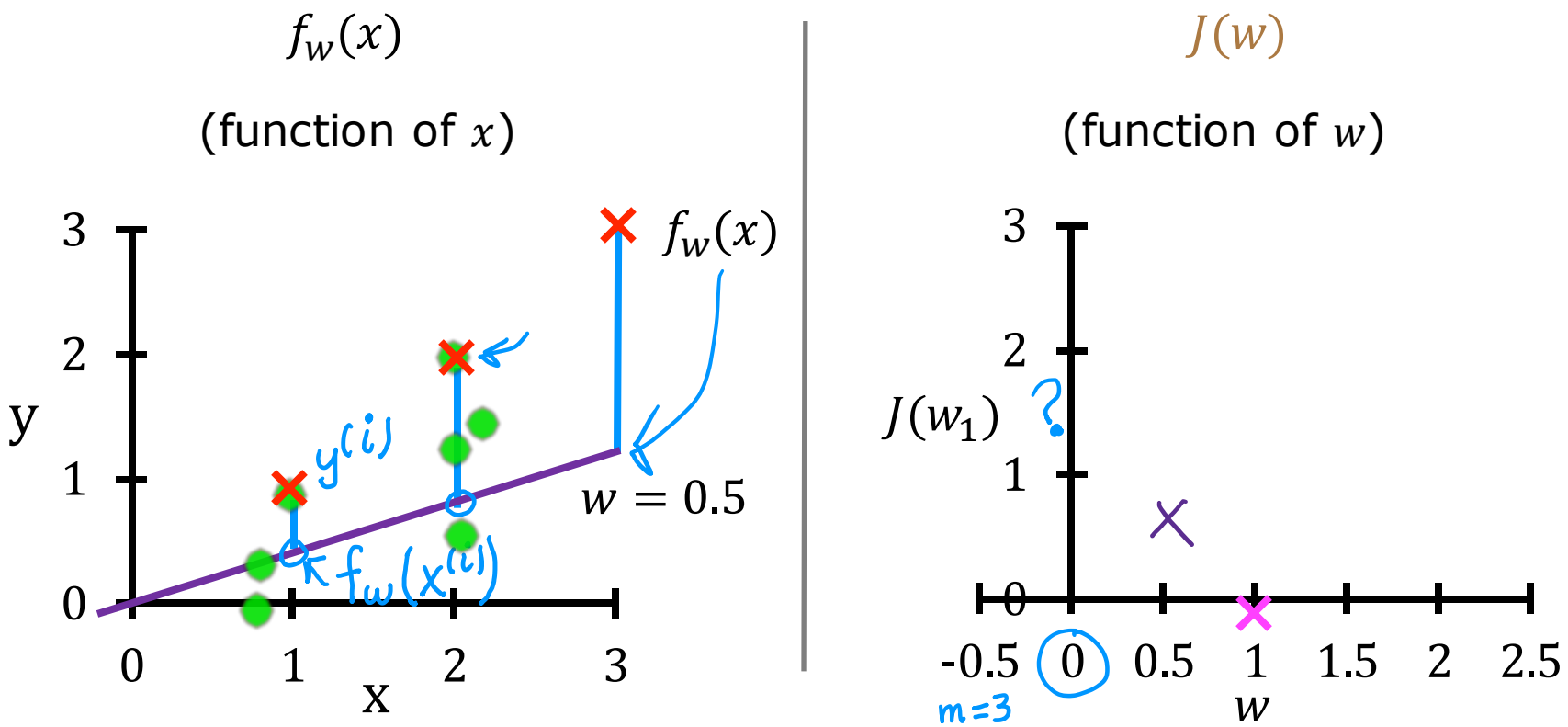
$wx^{(i)}$

$$\underset{w}{\text{minimize}}\, J(w)$$

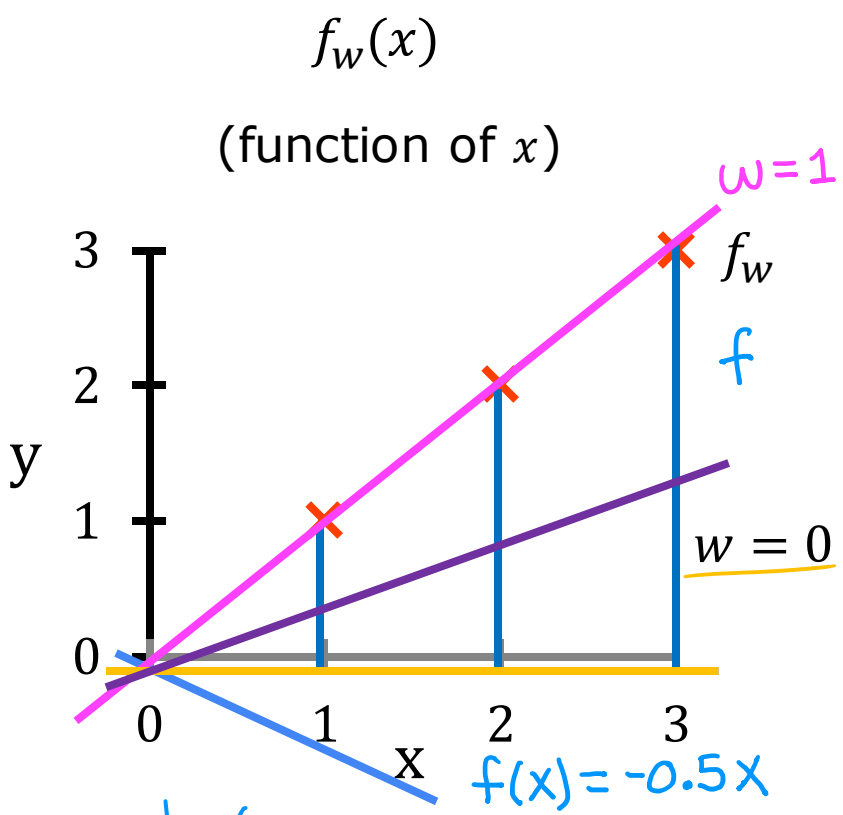


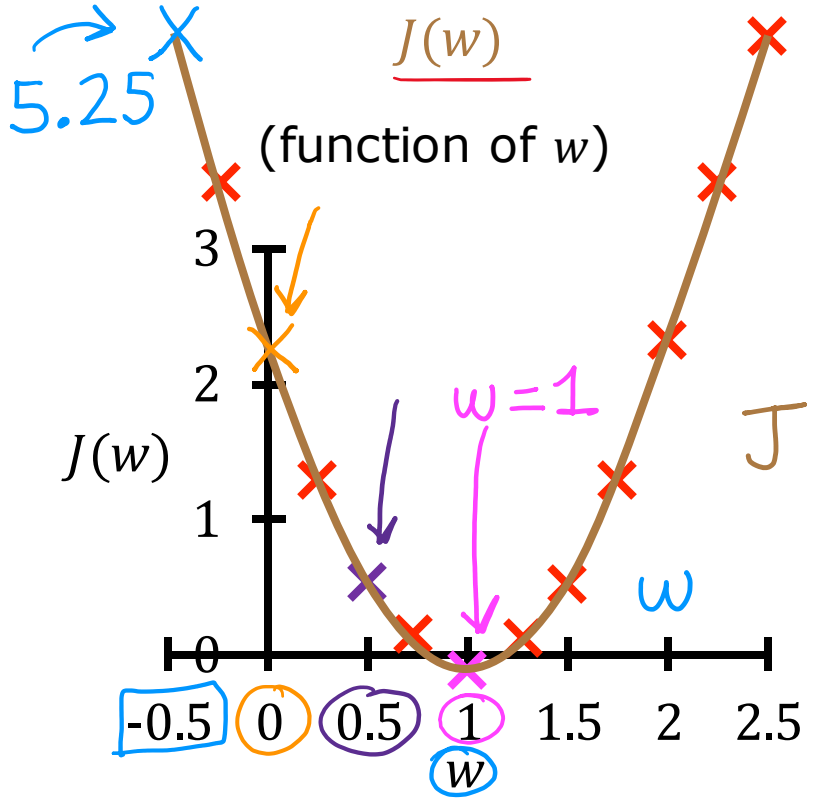


$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}(f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m}\sum_{i=1}^{m}(wx^{(i)} - y^{(i)})^2 = \frac{1}{2m}(0^2 + 0^2 + 0^2) = 0$$



$$J(0.5) = \frac{1}{2m}\left[(0.5-1)^2+(1-2)^2+(1.5-3)^2\right] = \frac{1}{2\times 3}[3.5] = \frac{3.5}{6} \approx 0.58$$





$$J(0) = \frac{1}{2m}\left(1^2+2^2+3^2\right) = \frac{1}{6}[14] \approx 2.3$$

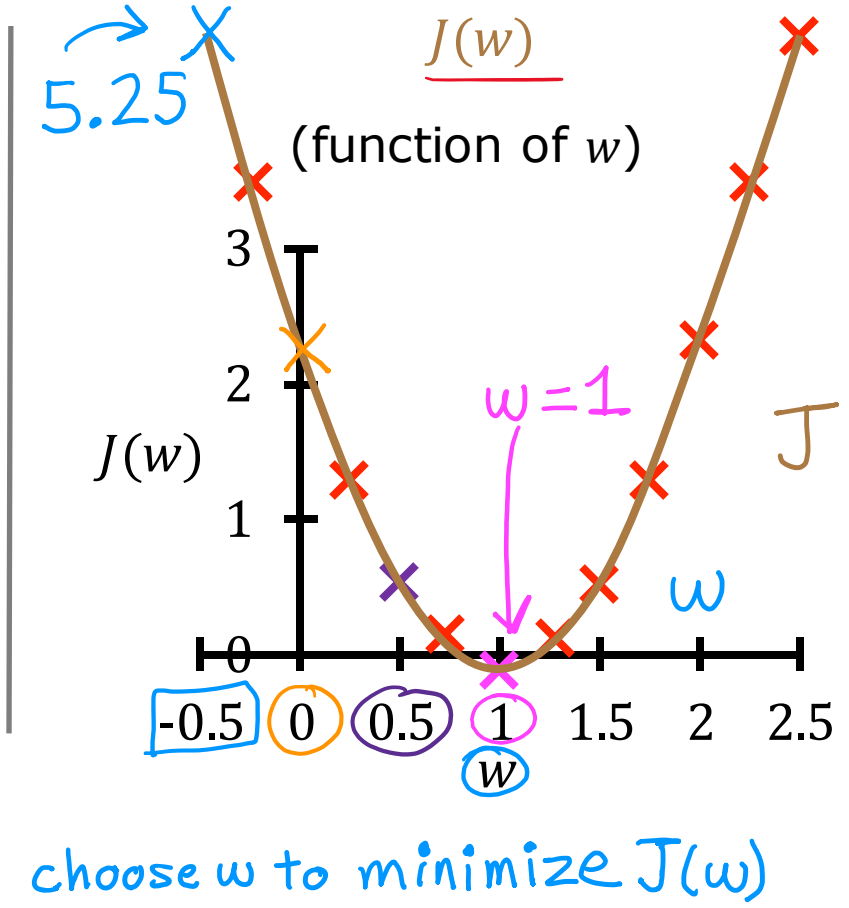
how to choose w?

goal of linear regression:

$$\underset{w}{\text{minimize}}\, J(w)$$

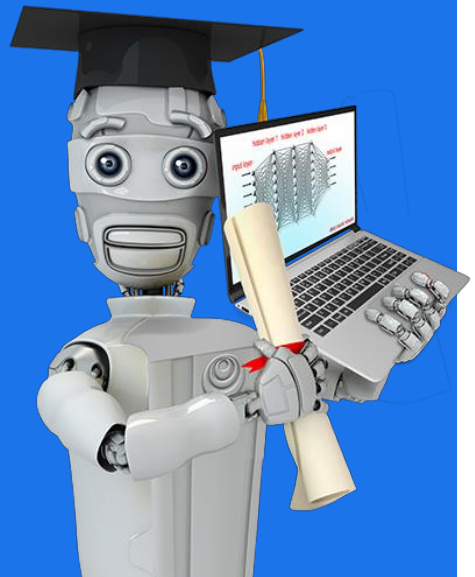
general case:

$$\underset{w,b}{\text{minimize}}\, J(w,b)$$



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Linear Regression with One Variable

Visualizing the Cost Function

Model $f_{w,b}(x) = wx + b$

~~before: b=0~~

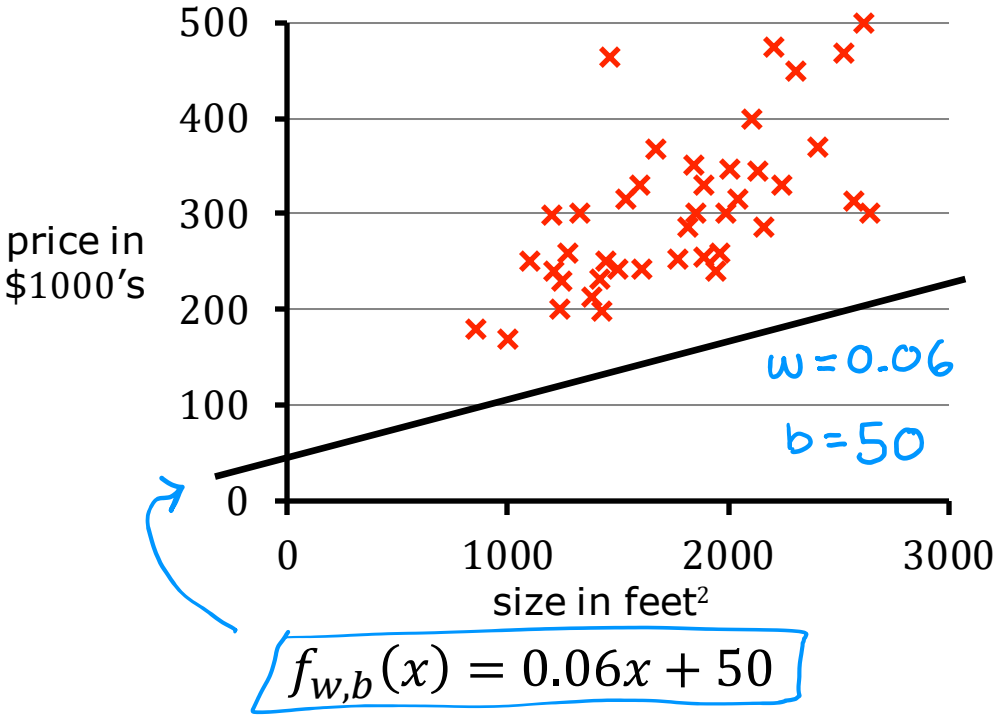
Parameters w, b

Cost Function $J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$

Objective $\underset{w,b}{\text{minimize}}\, J(w,b)$

$f_{w,b}$

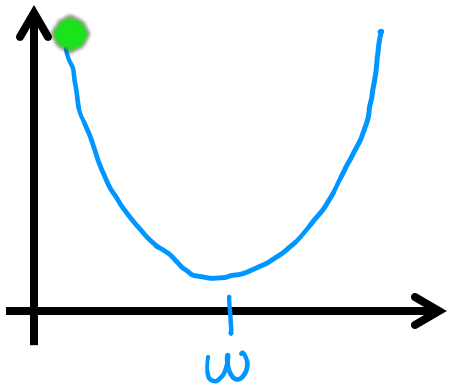
(function of x)



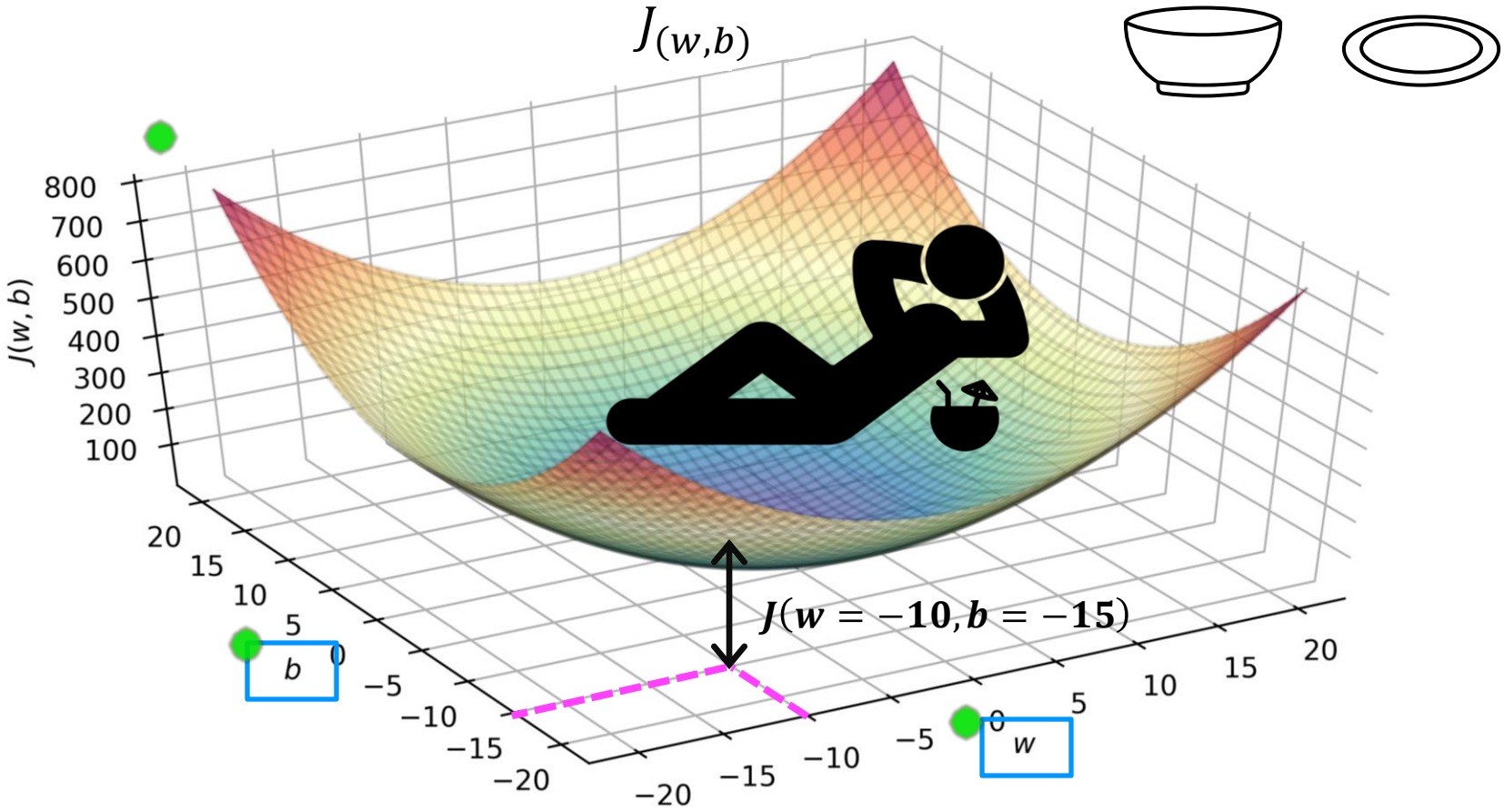
$f_{w,b}(x) = 0.06x + 50$

J

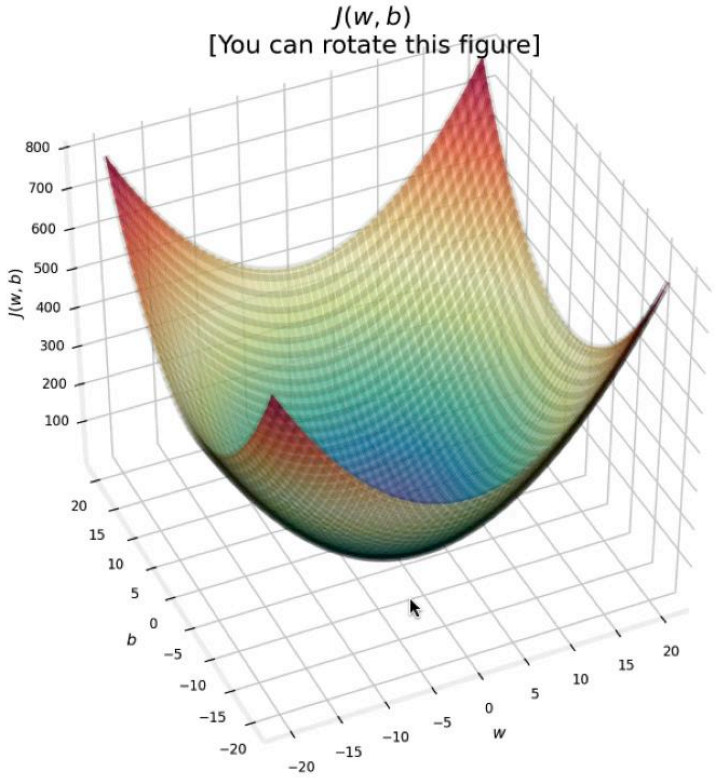
(function of w, b)



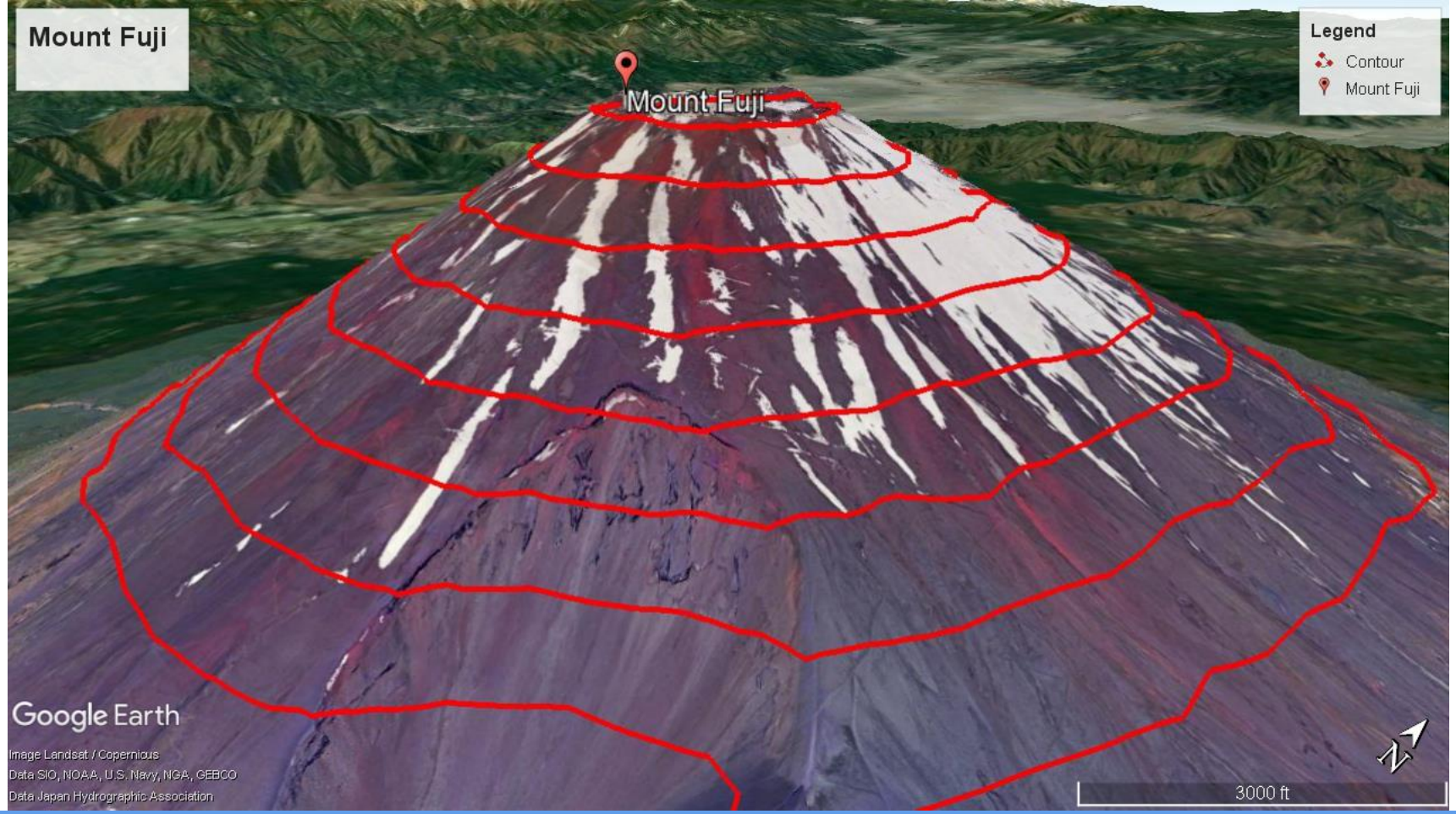
w , b

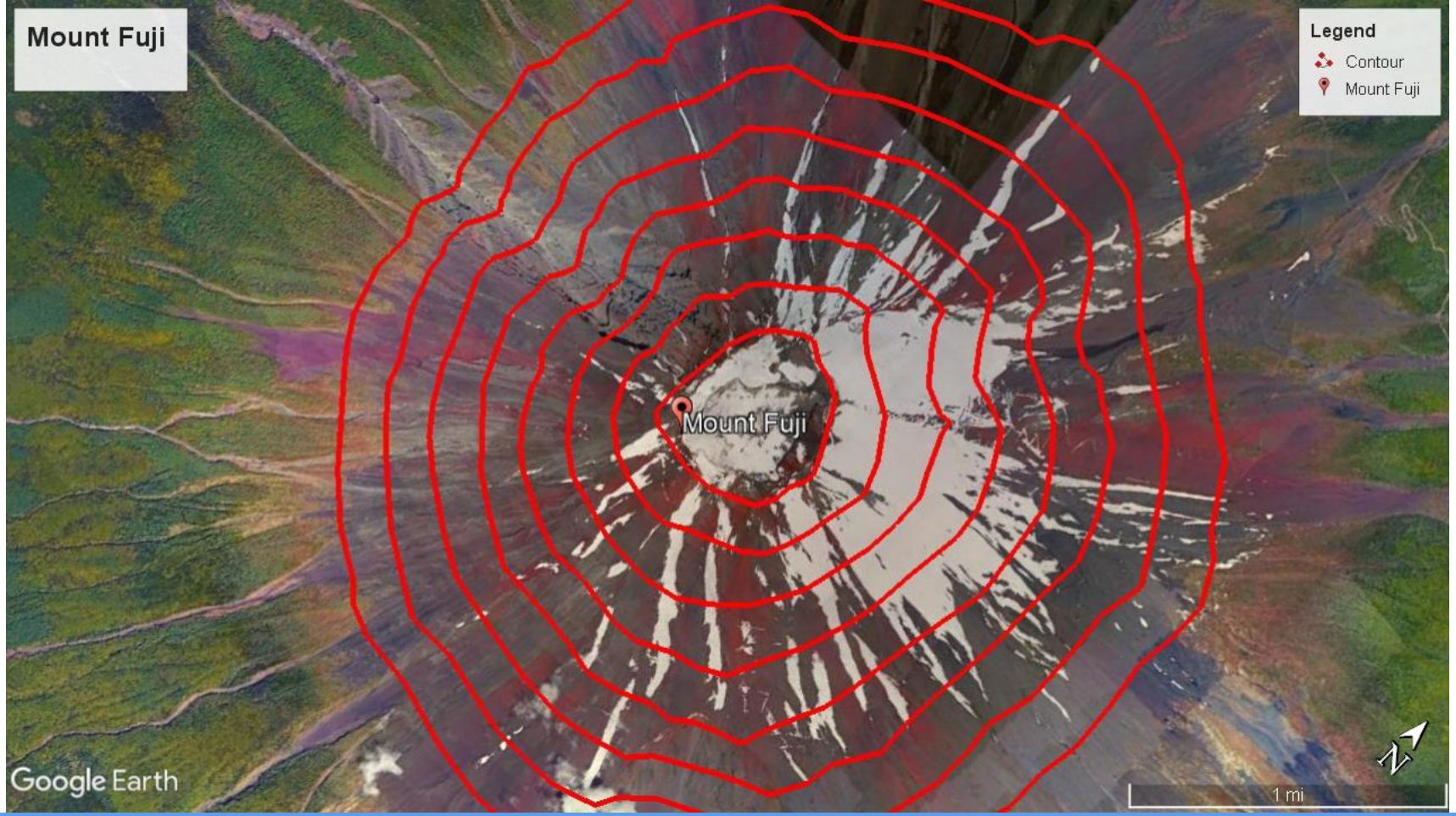

$J_{(w,b)}$
800
700
600
500
400
300
200
100
$J(w,b)$
20
15
10
5
0
−5
−10
−15
−20
b
$J(w=-10, b=-15)$
−20
−15
−10
−5
0
5
10
15
20
w

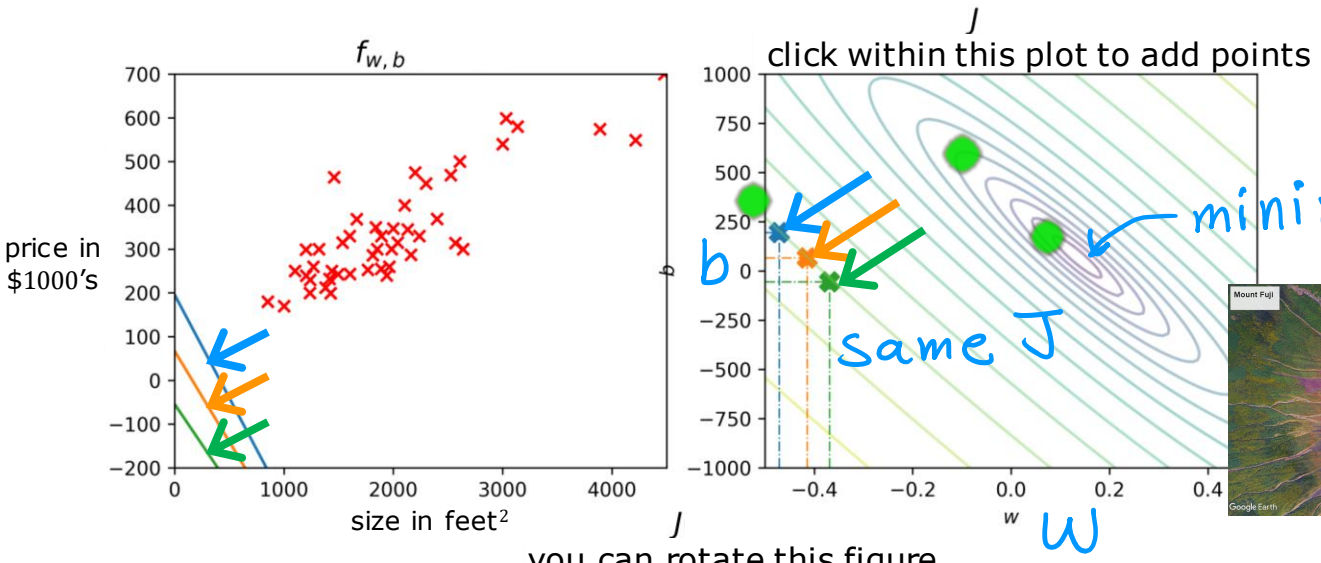
3D surface plot



Alternative contour plot

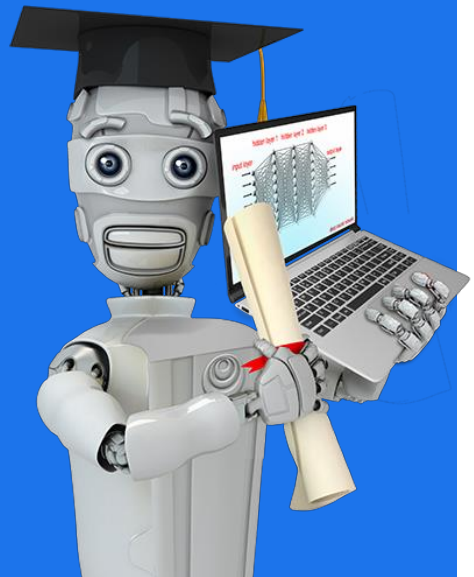

Mount Fuji
Legend
Contour
Mount Fuji
Mount Fuji
Google Earth
Image Landsat / Copernicus
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Data Japan Hydrographic Association
3000 ft


Mount Fuji
Legend
Contour
Mount Fuji
Mount Fuji
Google Earth
1 mi


$f_{w,b}$
price in $1000's
size in feet^2
J
click within this plot to add points
b
w
same J
minimum
J
you can rotate this figure
1e8
J(w, b)
minimum

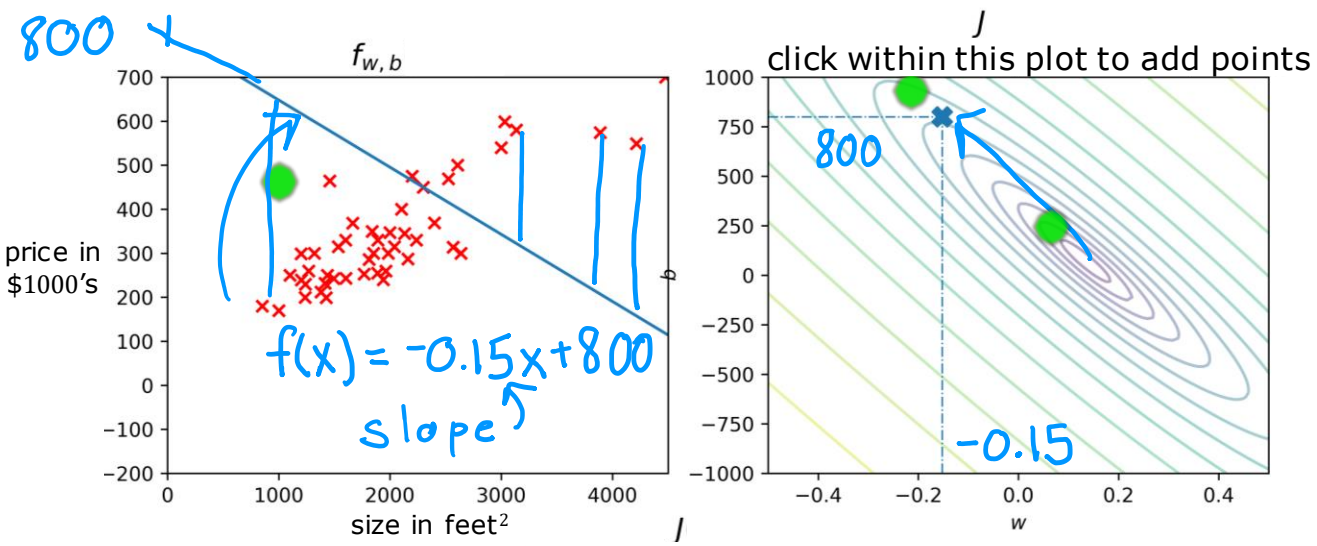
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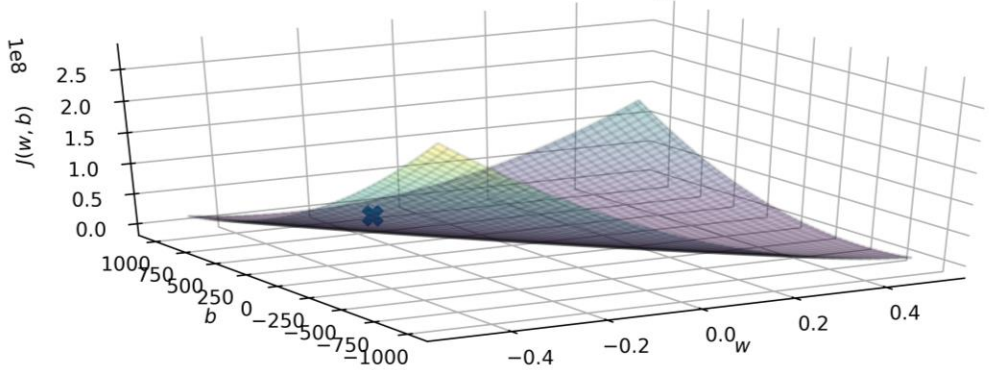


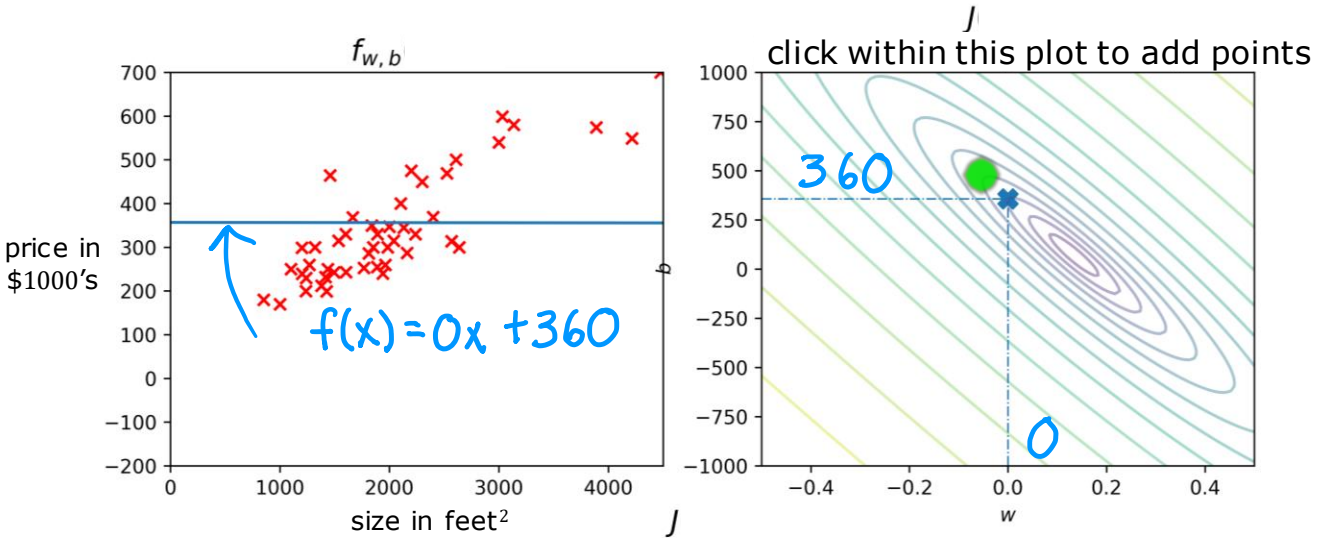
Linear Regression with One Variable

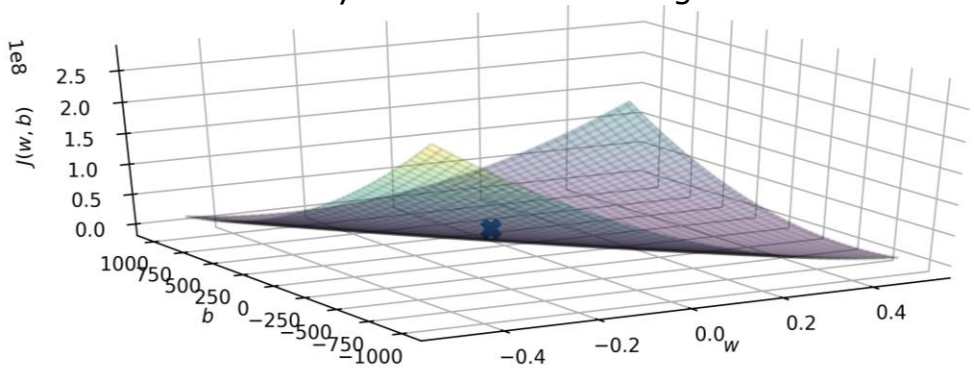
Visualization examples

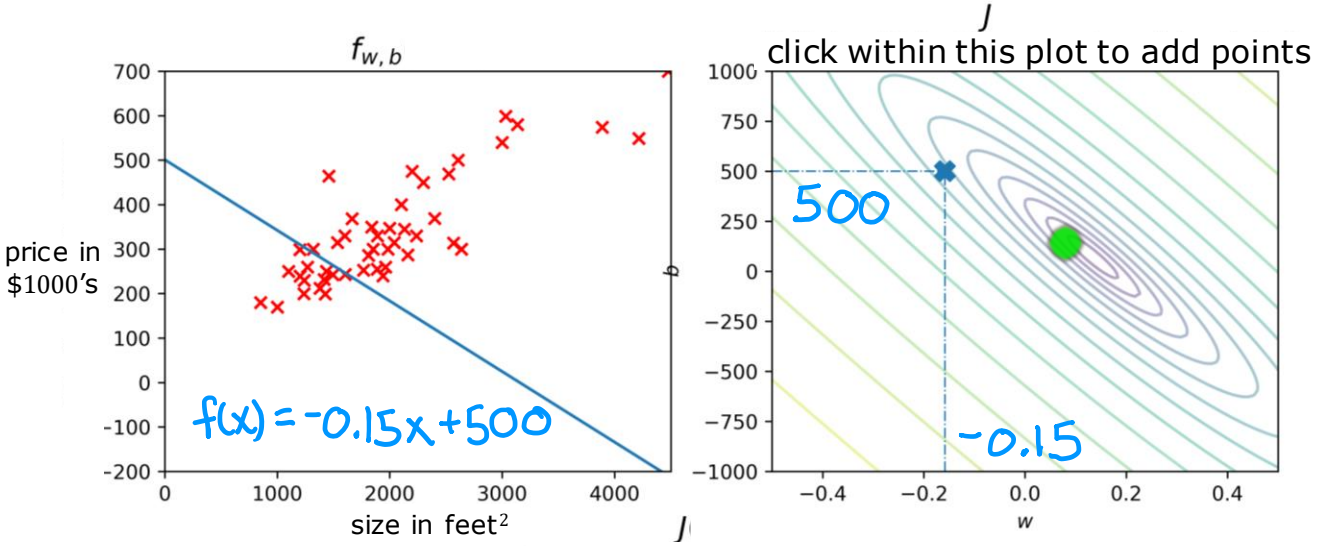


you can rotate this figure

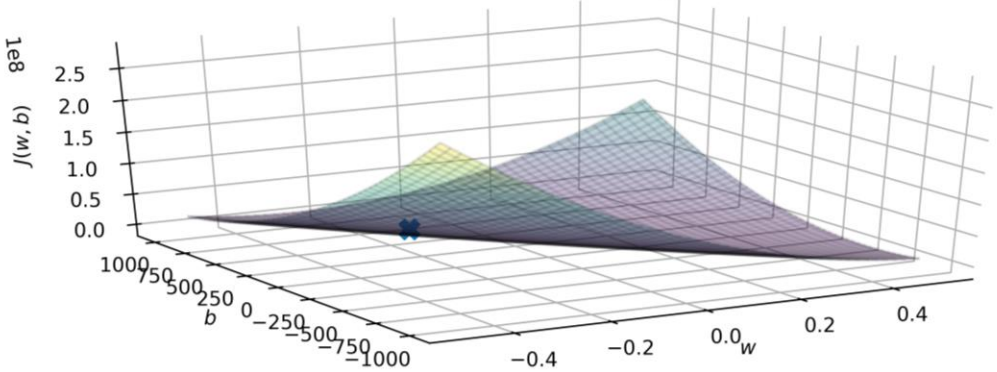


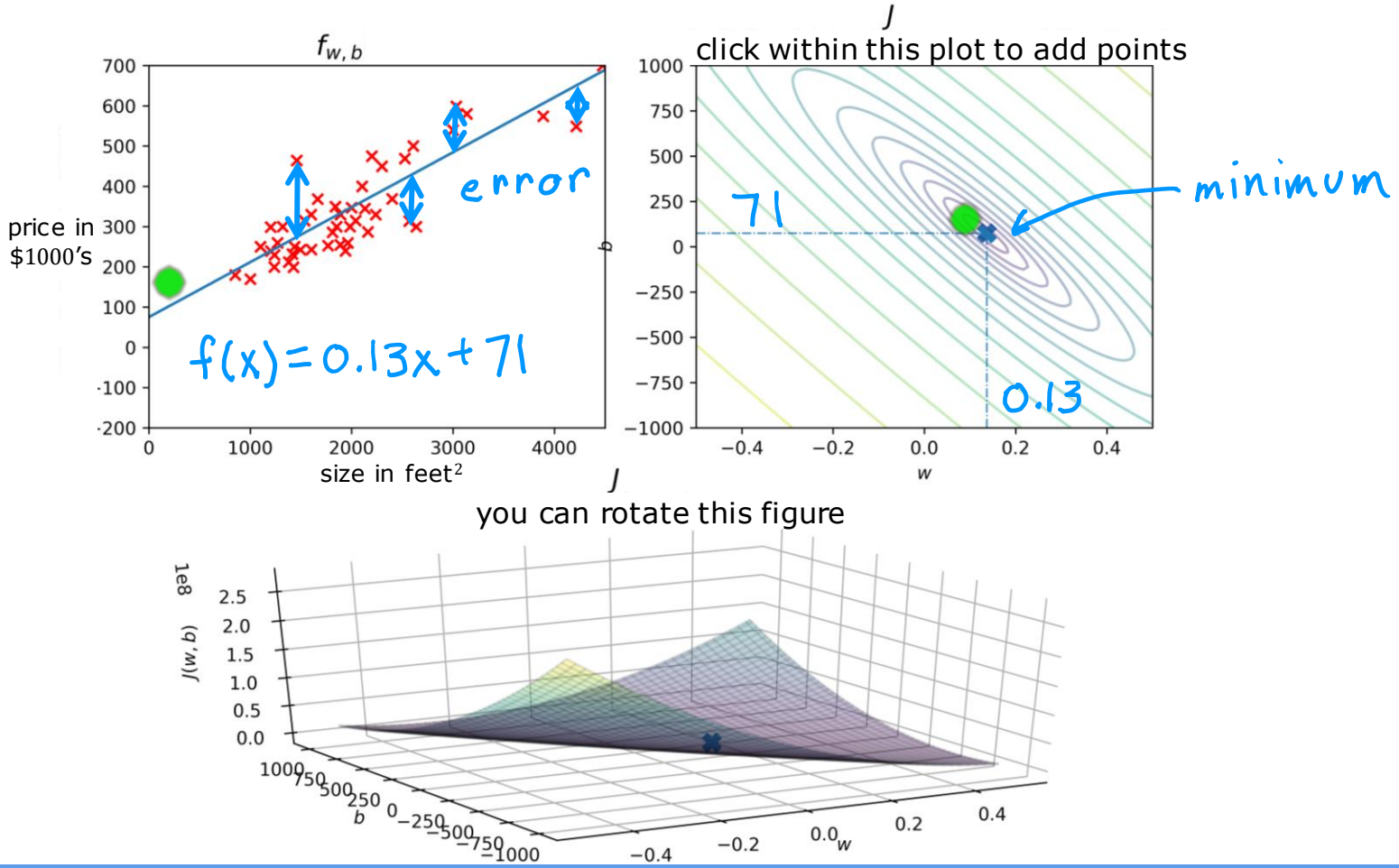

$f_{w,b}$
price in $1000's
size in feet^2
f(x) = 0x + 360
J
click within this plot to add points
360
0
b
w


J
you can rotate this figure
1e8
J(w, b)
b
w



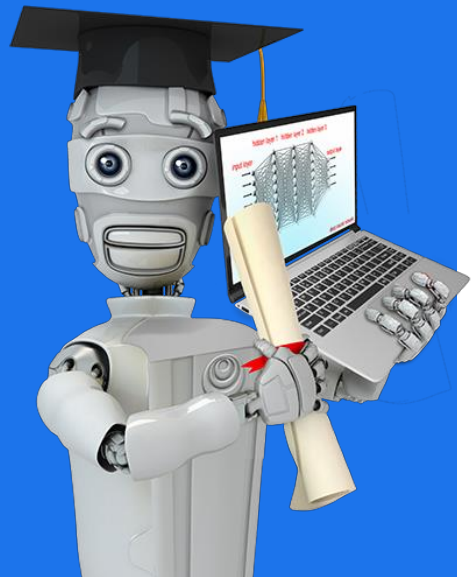
you can rotate this figure




$f_{w,b}$
price in $1000's
size in feet^2
error
f(x)=0.13x+71
J
click within this plot to add points
b
w
71
0.13
minimum
J
you can rotate this figure
1e8
J(w, b)
b
w

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Training Linear Regression

Gradient Descent

Have some function $J(w,b)$

for linear regression or any function

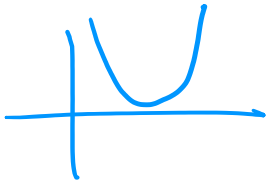
Want $\min_{w,b} J(w,b)$

$\min_{w_1, \ldots, w_n, b} J(w_1, w_2, \ldots, w_n, b)$

Outline:

- Start with some w, b (set $w=0, b=0$)
- Keep changing w, b to reduce $J(w,b)$
- Until we settle at or near a minimum

J not always



may have >1 minimum

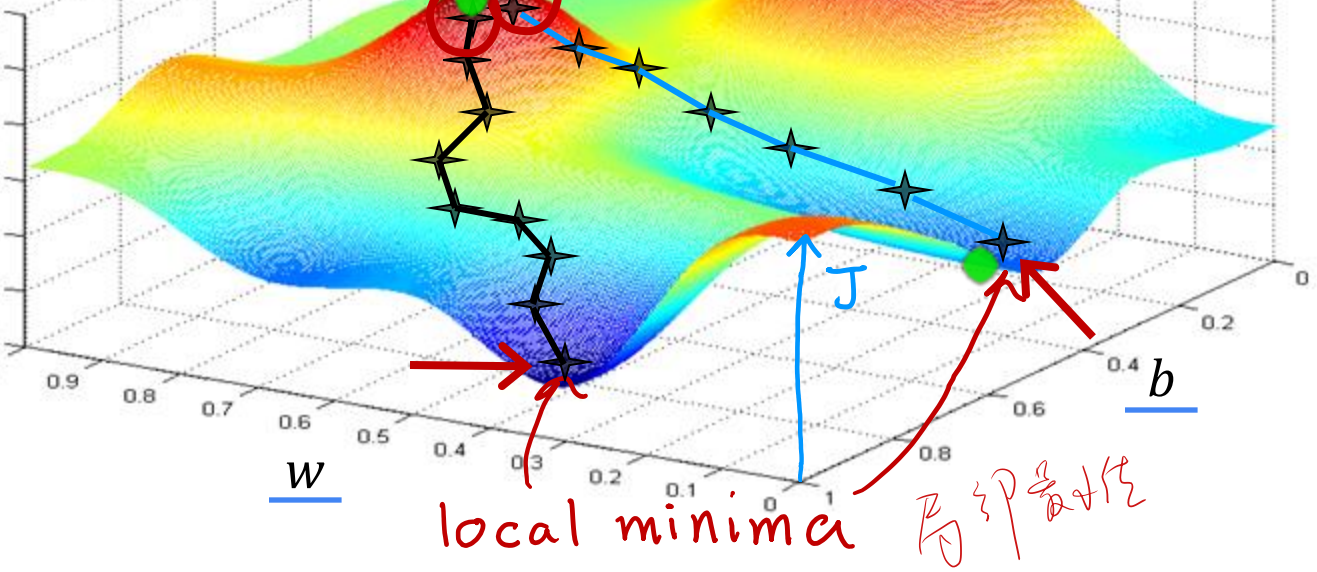
gradient descent



$J(w, b)$

not squared error cost

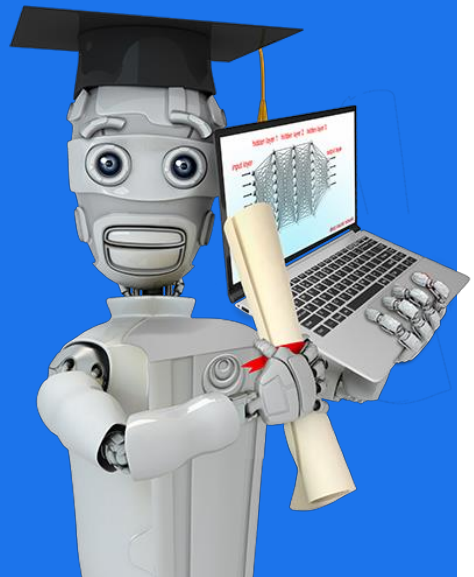
not linear regression



local minima

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Training Linear Regression

Implementing Gradient Descent

Gradient descent algorithm

Repeat until convergence

$$w = w - \alpha \frac{d}{dw} J(w,b)$$

$$b = b - \alpha \frac{d}{db} J(w,b)$$

Learning rate

Derivative 导数

Simultaneously update w and b

Assignment

$a = c$

$a = a + 1$

Code 赋值语句

Truth assertion

$a = c$

$a = a + 1$ ✗

Math

a==c

Correct: Simultaneous update

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$w = tmp_w$$

$$b = tmp_b$$

Incorrect

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$w = tmp_w$$

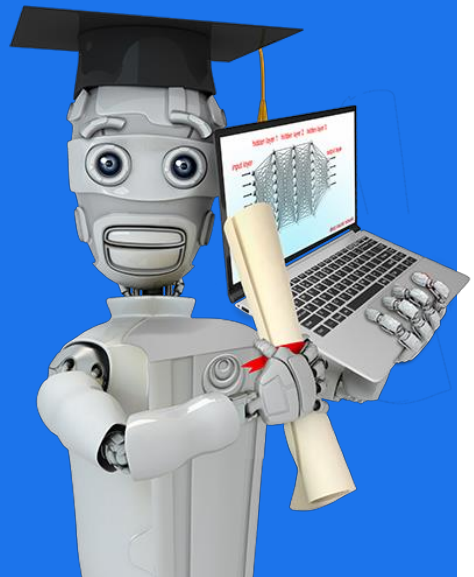
$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$b = tmp_b$$

用了未定义的 w

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Training Linear Regression

Gradient Descent Intuition

Gradient descent algorithm

repeat until convergence {

learning rate

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

derivative

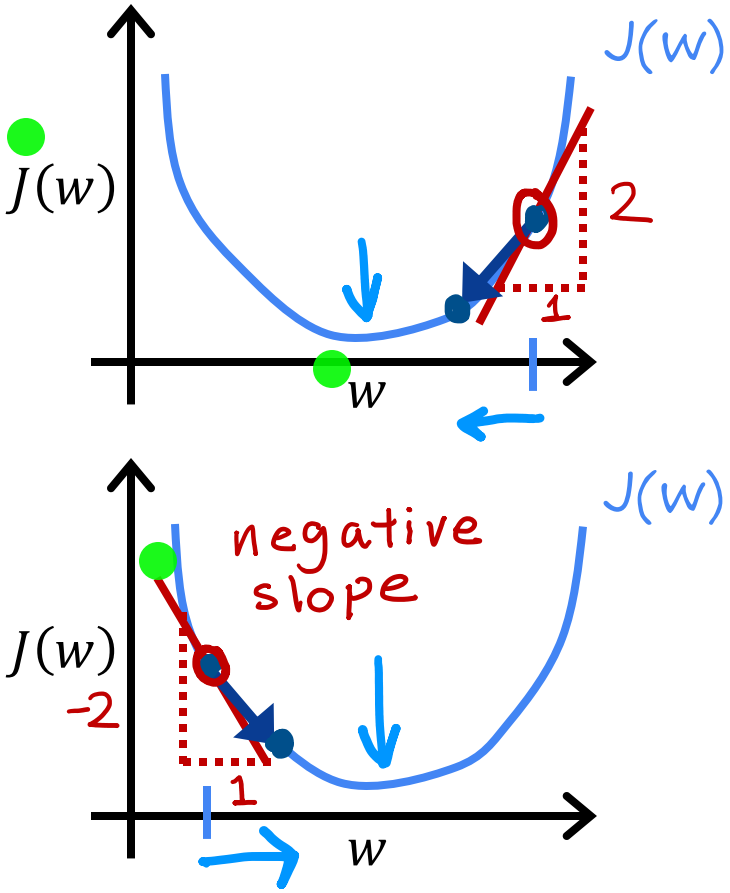
$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

}

$J(w)$

$$w = w - \alpha \frac{\partial}{\partial w} J(w)$$

$$\min_{w} J(w)$$



$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{>0}$$

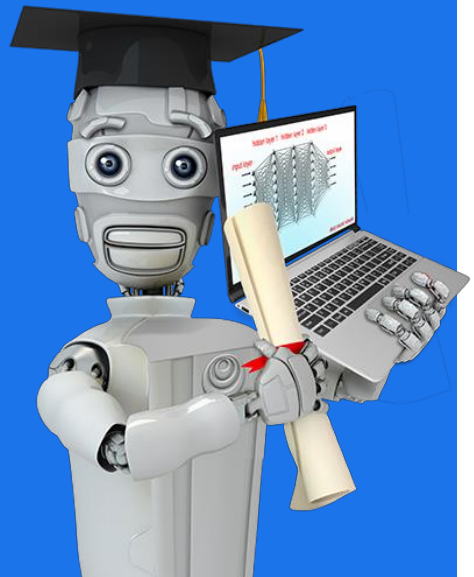
$$w = w - \alpha \cdot (positive\ number)$$

$$\underbrace{\frac{d}{dw} J(w)}_{<0}$$

$$w = w - \alpha \cdot (negative\ number)$$

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Training Linear Regression

Learning Rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

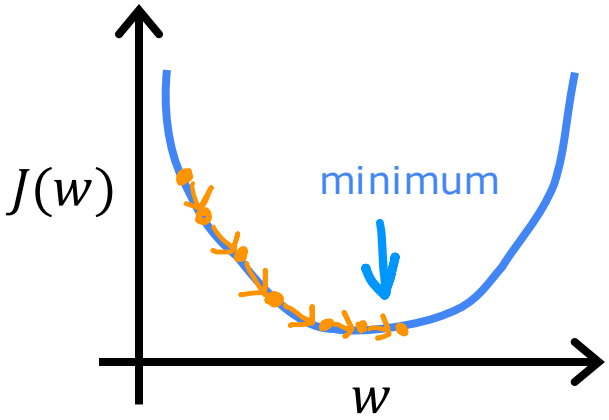
If α is too small…

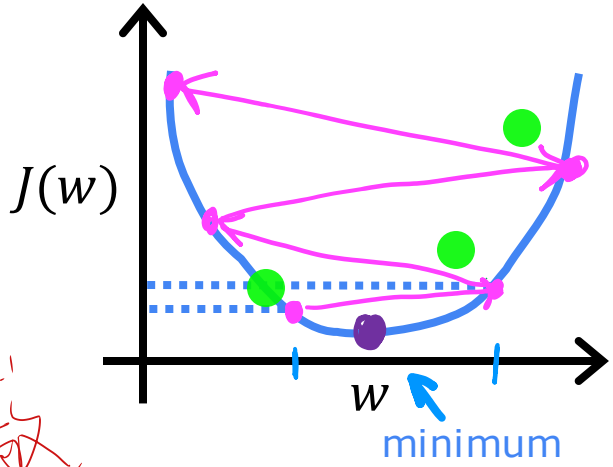
Gradient descent may be slow.

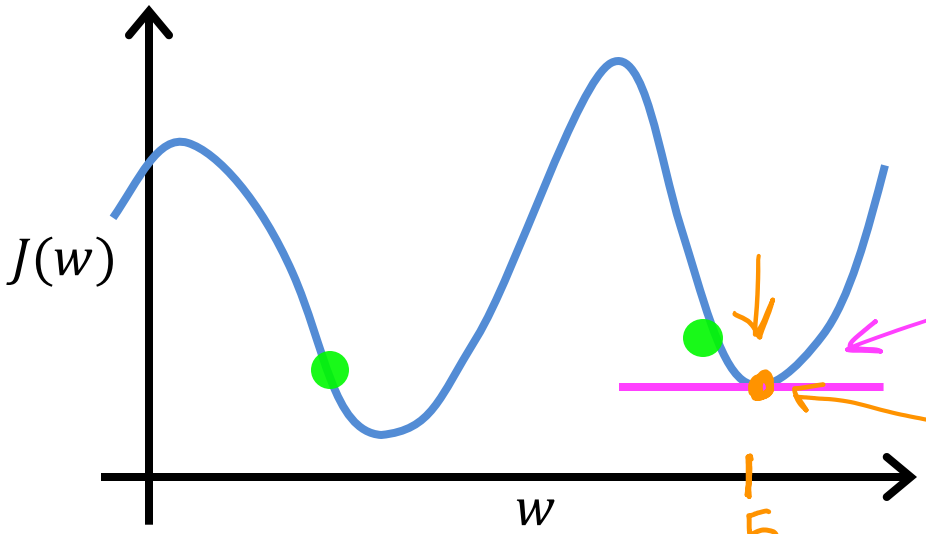
If α is too large…

Gradient descent may:

- Overshoot, never reach minimum
- Fail to converge, diverge







$$w = w - \alpha \frac{d}{dw} J(w)$$

0.1

$=0$

$5 - 0.1 \cdot 0$

$w = w - \alpha \cdot 0$

$w = w$

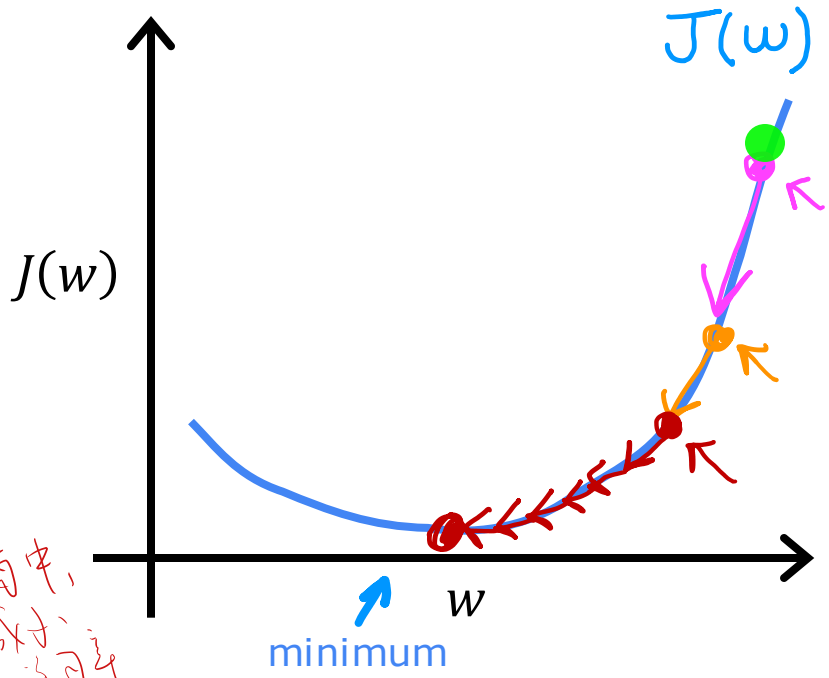
Can reach local minimum with fixed learning rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

smaller

not as large

large

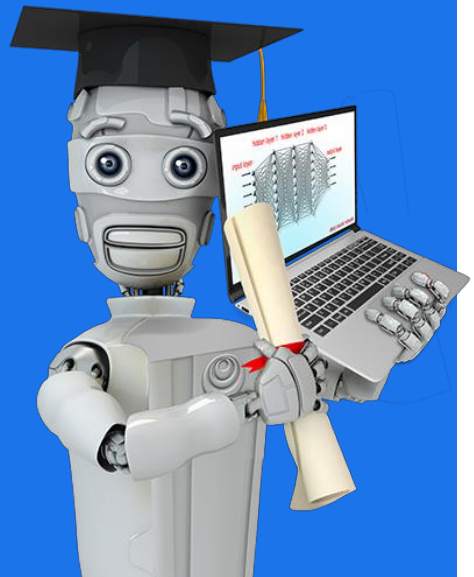


Near a local minimum,
- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate α

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Training Linear Regression

Gradient Descent for Linear Regression

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b) \rightarrow \frac{1}{m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})x^{(i)}$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b) \rightarrow \frac{1}{m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})$$

}

next slide is optional!

(Optional)

$$\frac{\partial}{\partial w} J(w,b) = \frac{d}{dw} \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{d}{dw} \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right)^2$$

$$= \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right) 2x^{(i)} = \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right) x^{(i)}$$

$$\frac{\partial}{\partial b} J(w,b) = \frac{d}{db} \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{d}{db} \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right)^2$$

$$= \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right) 2 = \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)$$

no $x^{(i)}$

Gradient descent algorithm

$\frac{\partial}{\partial w} J(w,b)$

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)} \right) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)} \right)$$

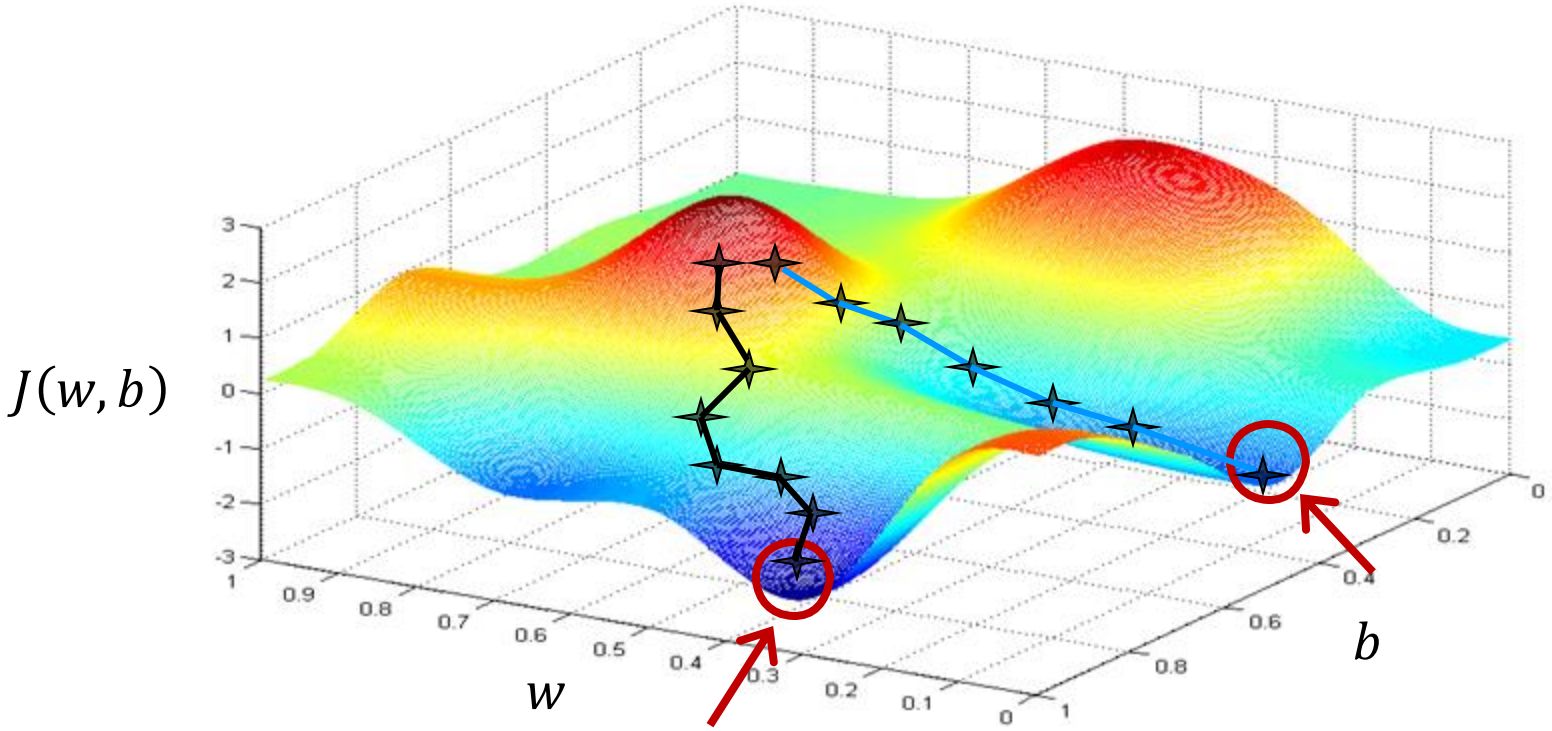
}

Update w and b simultaneously

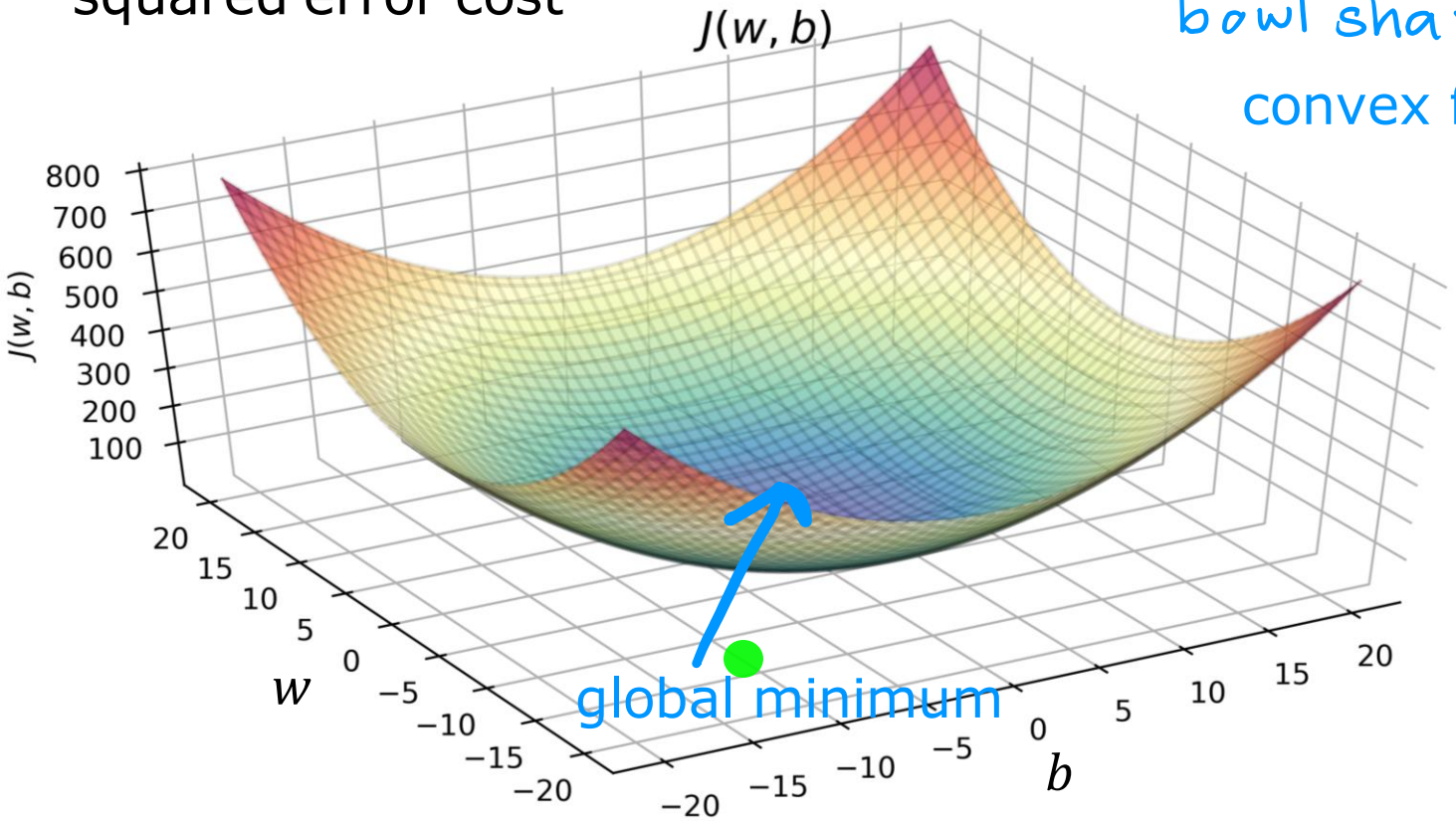
$\frac{\partial}{\partial b} J(w,b)$

$f_{w,b}(x^{(i)}) = wx^{(i)} + b$

More than one local minimum



squared error cost

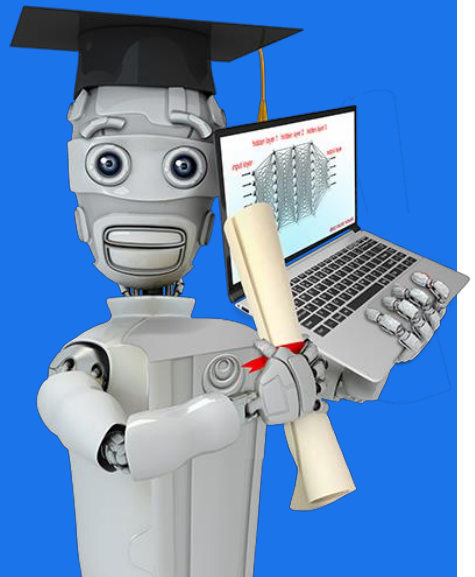


bowl shape

convex function

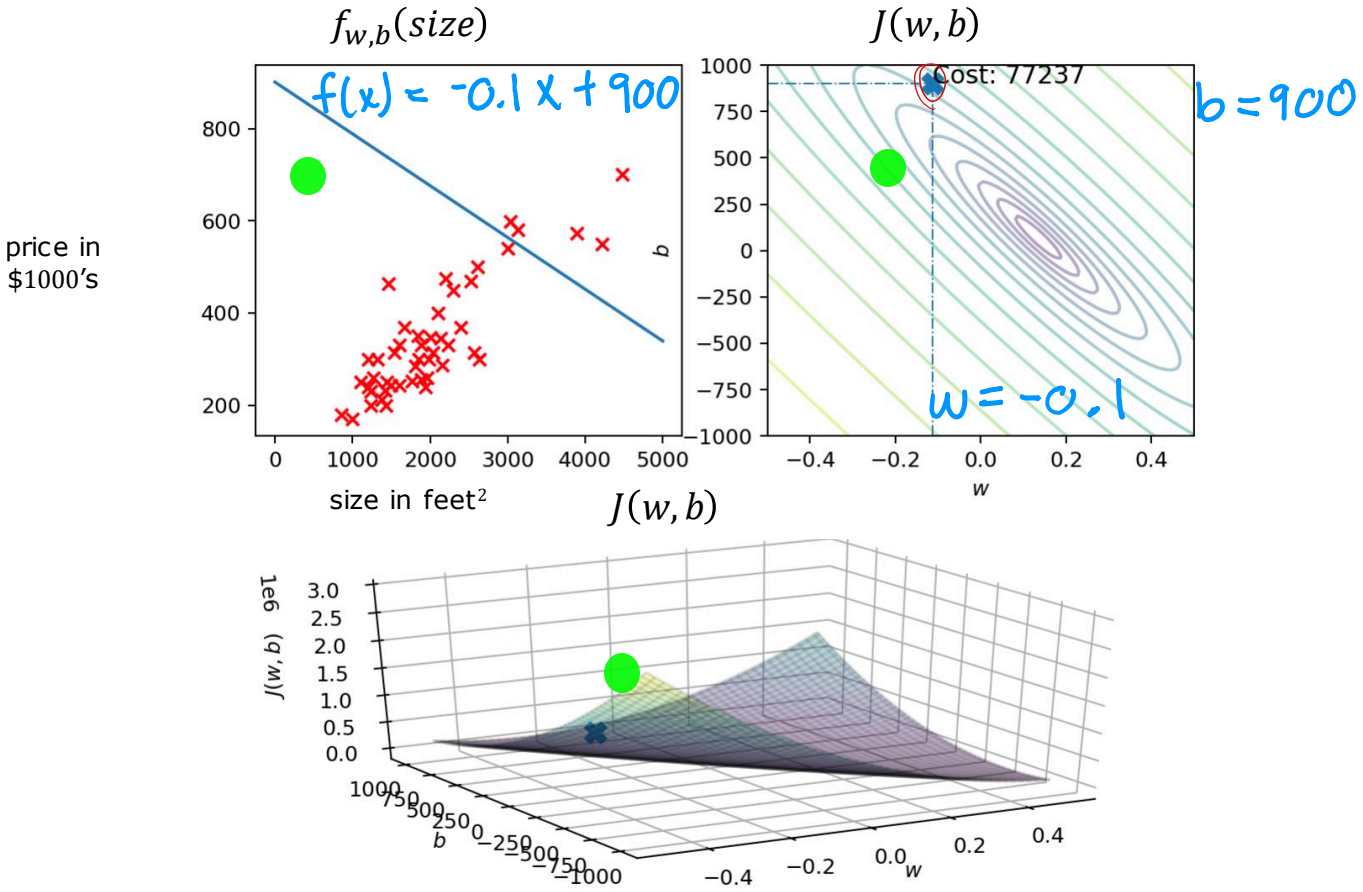
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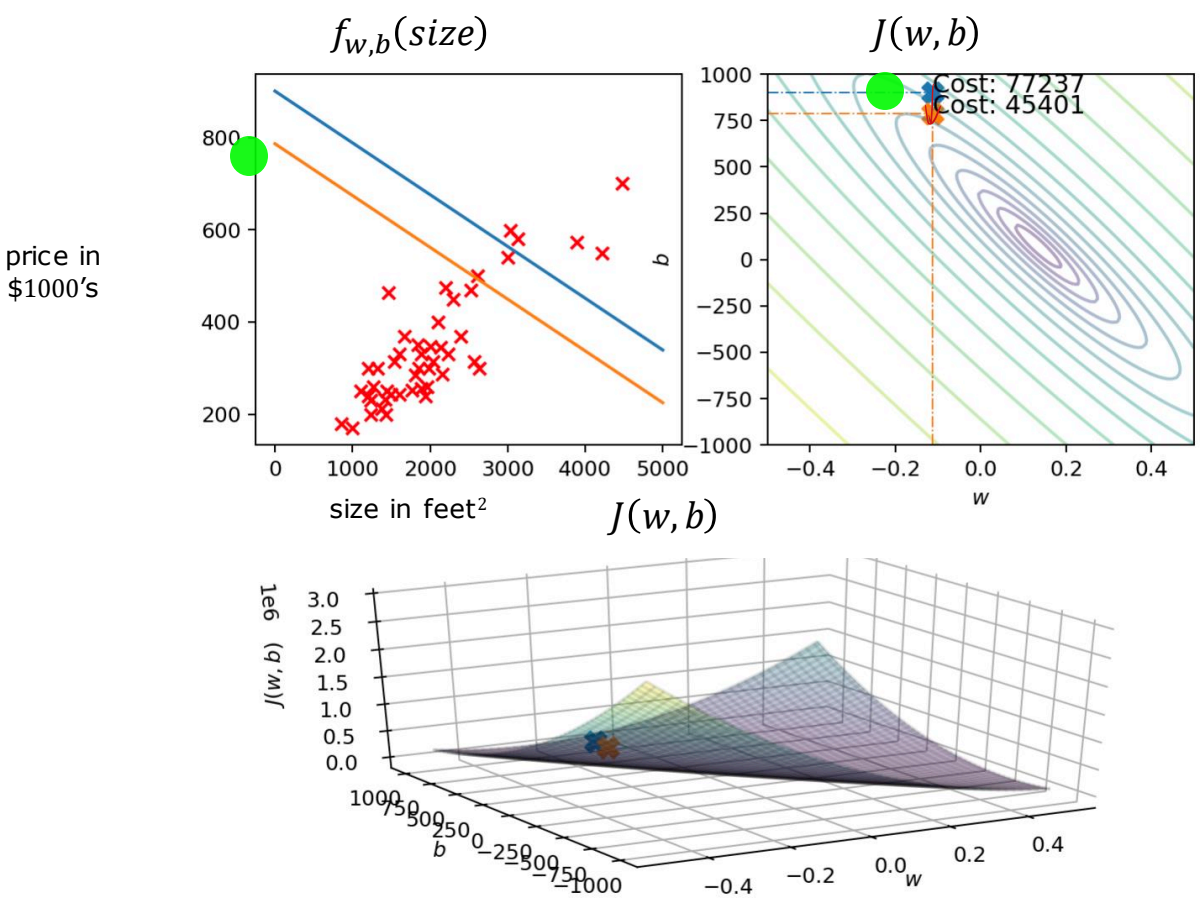
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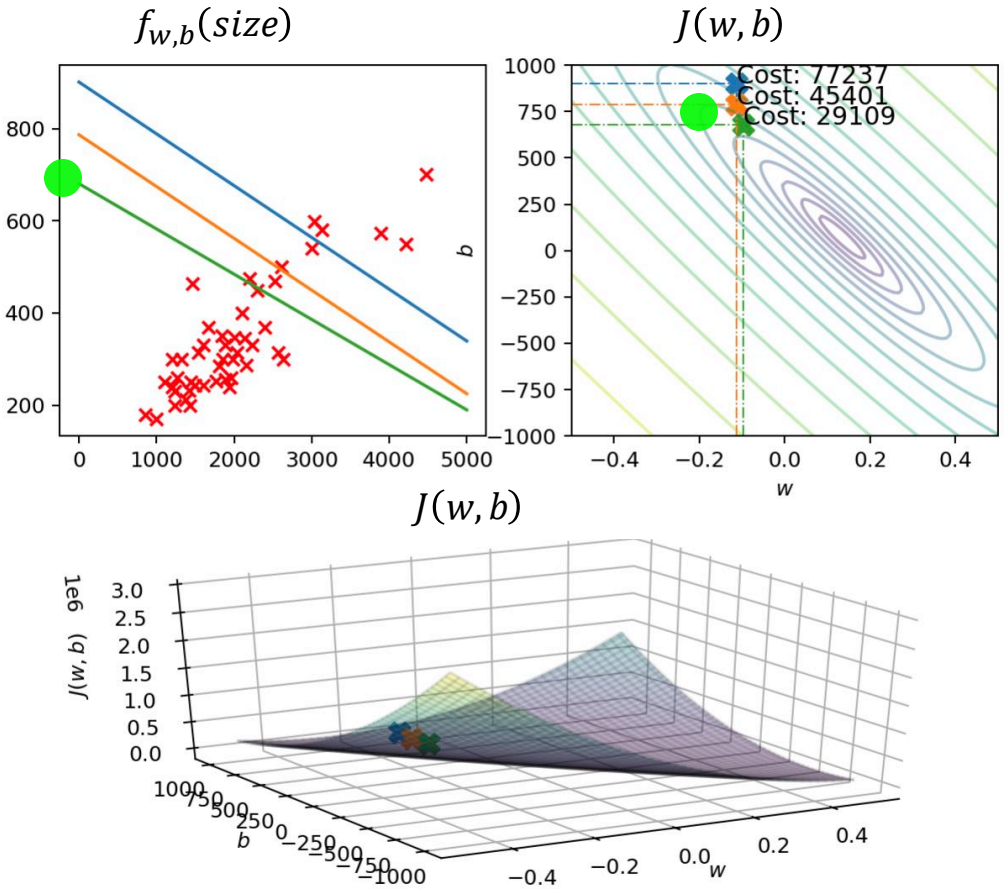


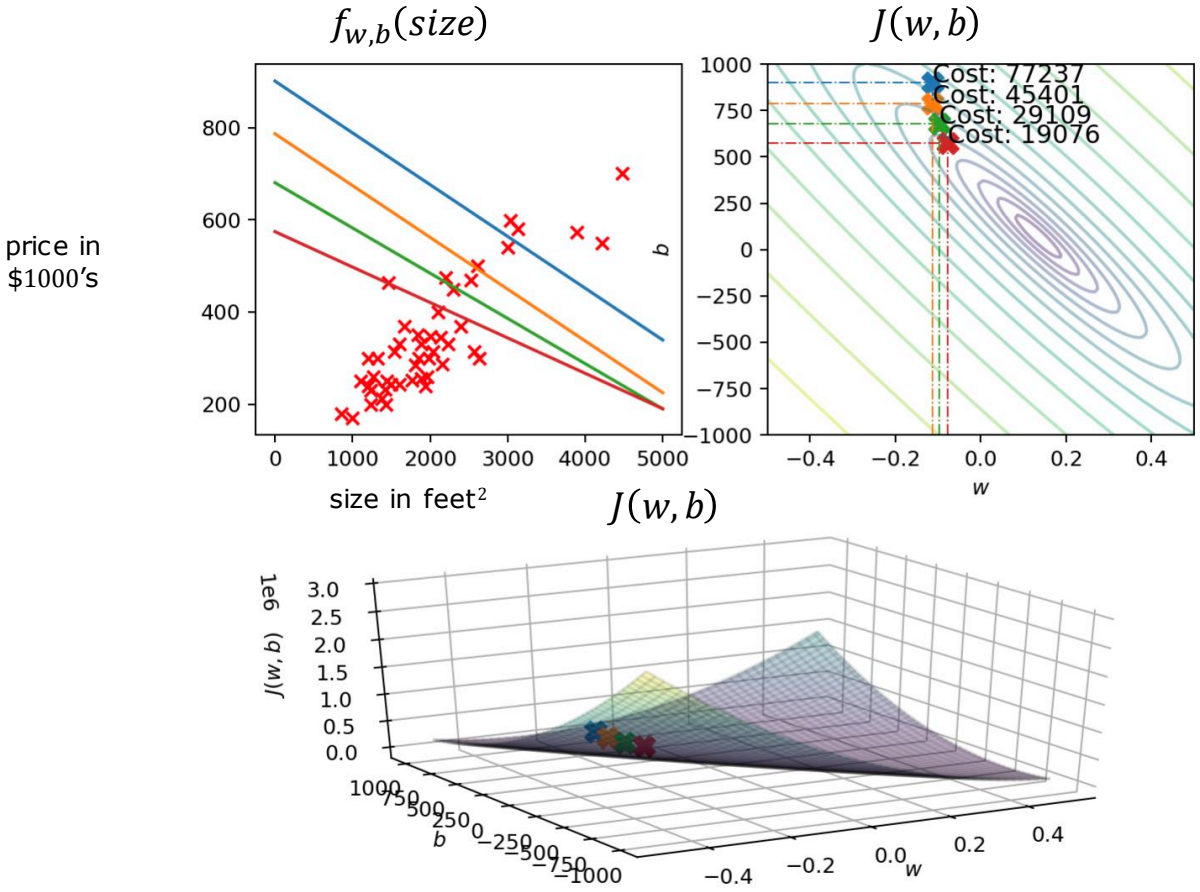
Training Linear Regression

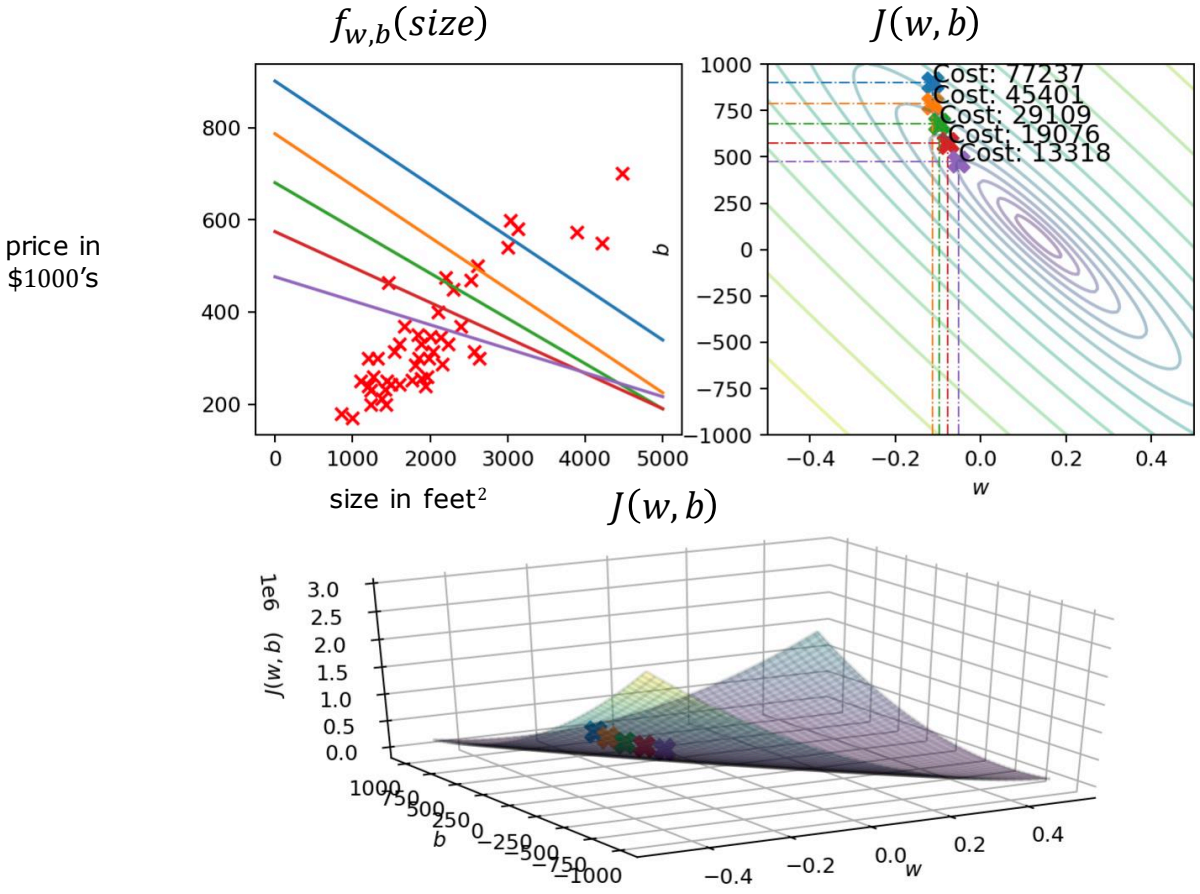
Running Gradient Descent

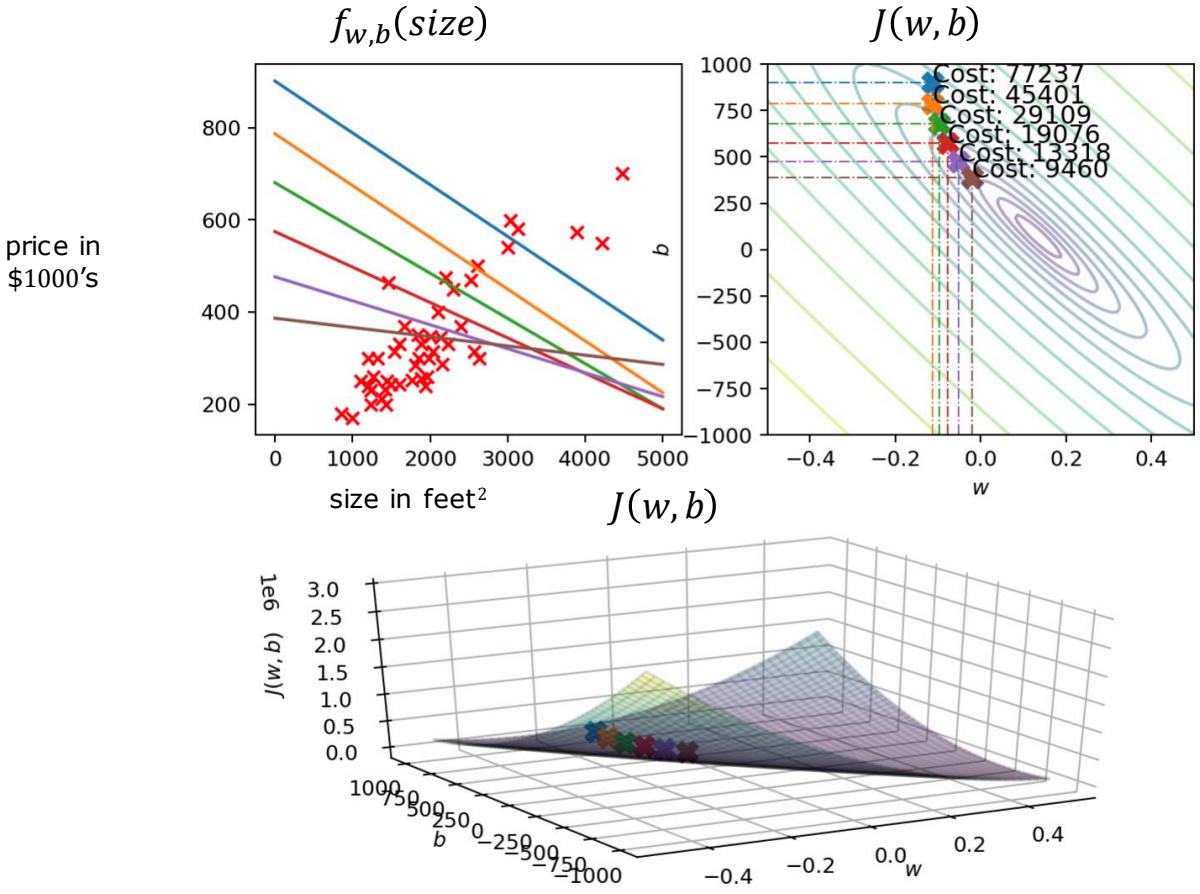

$f_{w,b}(size)$
$J(w,b)$
f(x) = -0.1 x + 900
Cost: 77237
b=900
w=-0.1
price in $1000's
size in feet²
$J(w,b)$
$J(w,b)$
b
w

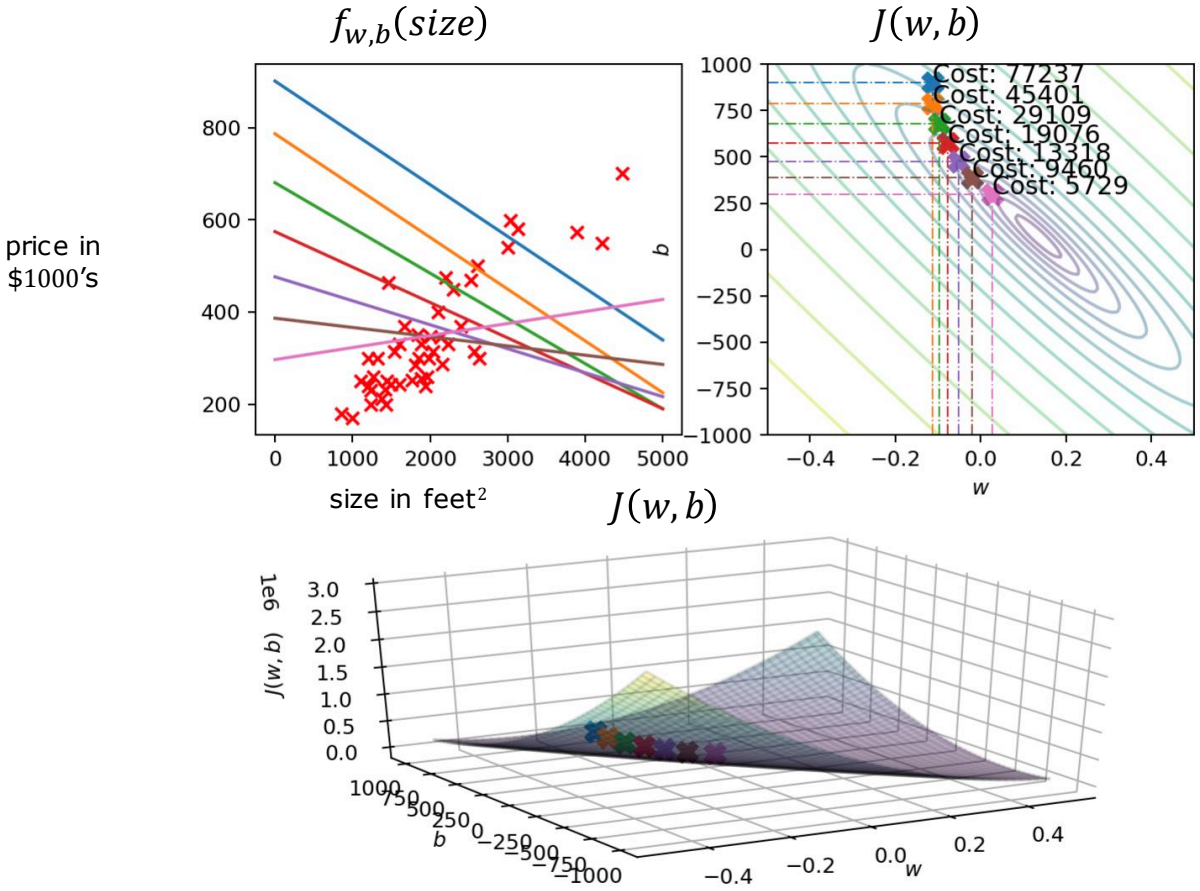

$f_{w,b}(size)$
$J(w,b)$
price in $1000's
size in feet^2
Cost: 77237
Cost: 45401
$J(w,b)$
log
$J(w,b)$
b
w

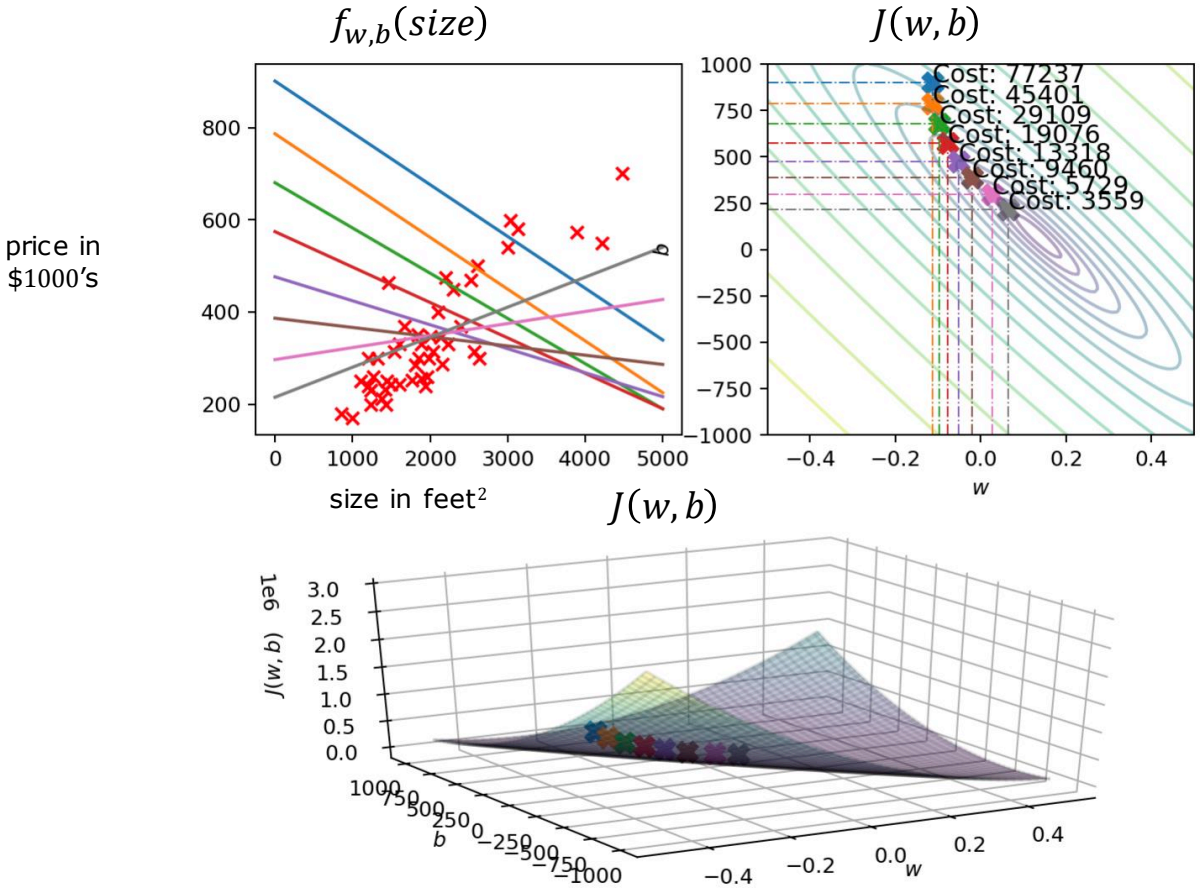

$f_{w,b}(size)$
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
b
w
$J(w,b)$
$J(w,b)$
log
b
w

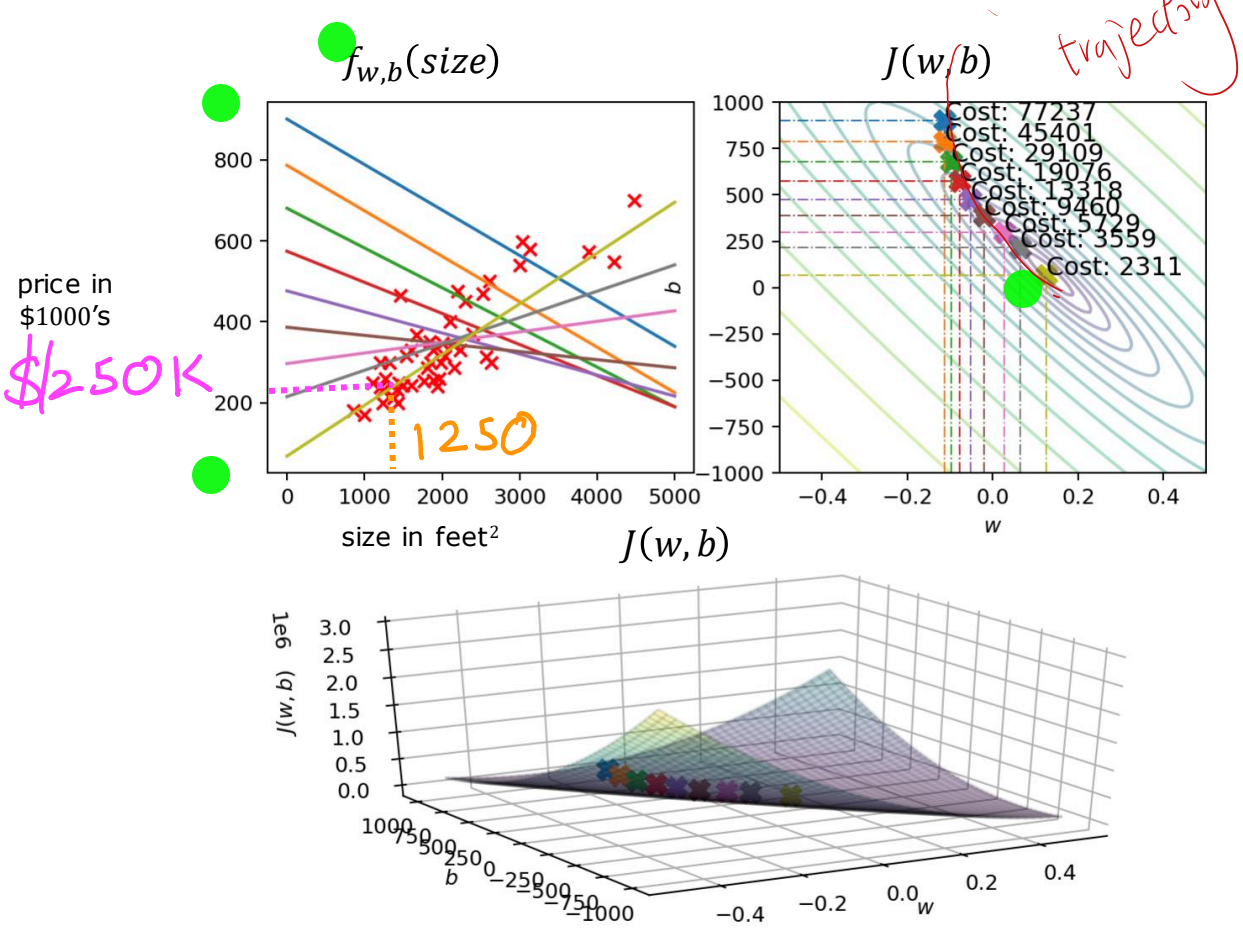

$f_{w,b}(size)$
price in $1000's
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
w
b
$J(w,b)$
log
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
size in feet^2
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
size in feet^2
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
$J(w,b)$
1e6


$f_{w,b}(size)$
price in $1000's
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
Cost: 5729
w
b
$J(w,b)$
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
Cost: 5729
Cost: 3559
$J(w,b)$
$J(w,b)$
w
b


$f_{w,b}(size)$
price in $1000's
$250K
1250
size in feet²
$J(w,b)$
trajectory
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
Cost: 5729
Cost: 3559
Cost: 2311
b
w
$J(w,b)$
J(w, b)
b
w

"Batch" gradient descent

DeepLearning.AI
THE BATCH

"Batch": Each step of gradient descent uses all the training examples.

other gradient descent: subsets

	x size in feet^2	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$\sum_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

